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Harmonic gear device and actuator

Abstract

A harmonic gear device includes a rigid internal gear, a flexible external gear and a wave generator. The rigid internal gear is an annular component having internal teeth. The flexible external gear is an annular component, which has external teeth and is configured on an inner side of the rigid internal gear. The wave generator is configured on an inner side of the flexible external gear, and deflects the flexible external gear. The harmonic gear device deforms the flexible external gear along with the rotation of the wave generator taking a rotation axis as the center, such that some of the external teeth mesh with some of the internal teeth. The flexible external gear rotates relative to the rigid internal gear in accordance with the difference between the numbers of teeth of the flexible external gear and the rigid internal gear.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of PCT International Application No. PCT/CN2021/083685, filed on Mar. 29, 2021, which claims the benefit of Japanese Patent Application No. 2020-174376, filed on Oct. 16, 2020, the contents of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD
(1) Embodiments of the disclosure generally relate to a harmonic gear device and an actuator, and in particular to a harmonic gear device having a rigid internal gear, a flexible external gear and a wave generator, and an actuator.
BACKGROUND
(2) A flexible external gear in a harmonic gear device (a flexural meshing type gear device) may have a surface treatment by way of nitridation.
(3) The harmonic gear device has: a rigid internal gear with an annular shape; a flexible external gear with a cup-shape, arranged on an inner side of the rigid internal gear; and a wave generator with an elliptical shape, embedded on an inner side of the flexible internal gear. The flexible external gear includes a trunk portion with a cylindrical shape and external teeth formed on an outer peripheral surface of the trunk portion. The flexible external gear is deflected into an elliptical shape by the wave generator, and a part of external teeth located at both end portions in a long-axis direction of the elliptical shape is meshed with internal teeth formed on an inner peripheral surface of the rigid internal gear.
(4) When the wave generator is rotated by a motor or the like, a meshing position of two gears moves in a circumferential direction, and a relative rotation corresponding to a difference (2N (N is a positive integer)) between tooth numbers of the internal teeth and the external teeth is generated between the two gears. Here, when a side of the rigid internal gear is fixed, a rotational output may be obtained from a side of the flexible external gear that may be greatly decelerated by an amount that corresponds to a difference between tooth numbers of the two gears,
SUMMARY
(5) However, the harmonic gear device deflects the flexible external gear while transmitting power by meshing of the internal teeth and the external teeth, therefore especially when it is used for a long period of time, a foreign matter such as metal powder, or nitride, or the like may be generated for example due to absence, or wear, or the like induced by contact between the internal teeth and the external teeth. Due to generation of such foreign matter, it is possible for the foreign matter to become engaged between the internal tooth and the external tooth or for the foreign matter to enter a bearing of the wave generator to cause damage to the bearing, which may affect reliability of the harmonic gear device.
(6) Some embodiments of the disclosure may provide a harmonic gear device and an actuator with high reliability.
(7) A harmonic gear device according to an aspect of some embodiments of the disclosure includes a rigid internal gear, a flexible external gear and a wave generator. The rigid internal gear is an annular component having a first number of internal teeth. The flexible external gear is an annular

component having a second number of external teeth and arranged on an inner side of the rigid internal gear. The wave generator is arranged on an inner side of the flexible external gear and configured to deflect the flexible external gear. The harmonic gear device is configured to deform the flexible external gear by a rotation of the wave generator along a rotation axis, such that a portion of the external teeth is meshed with a portion of the internal teeth, and the flexible external gear is configured to rotate relative to the rigid internal gear in accordance with a difference between the first number of internal teeth and the second number of external teeth. An internal tooth of the first number of internal teeth has a tooth-direction trimming portion at an end portion in at least one side along a tooth direction of the internal tooth.
(8) An actuator according to an aspect of some embodiments of the disclosure includes the above harmonic gear device, a driving source and an output portion. The driving source is configured to rotate the wave generator. The output portion is configured to provide a rotation force of one of the rigid internal gear or the flexible external gear out as an output.
(9) According to some embodiments of the disclosure, a harmonic gear device and an actuator with high reliability may be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a cross-sectional view showing a general structure of a harmonic gear device according to some embodiments.
(2) FIG. **1**B is an enlarged view of a region Z**1** of FIG. **1**A.
(3) FIG. **2** is a diagrammatic view of the above harmonic gear device observed from an input side of a rotation axis.
(4) FIG. **3**A is a diagrammatic exploded perspective view of the above harmonic gear device observed from an output side of the rotation axis.
(5) FIG. **3**B is a diagrammatic exploded perspective view of the above harmonic gear device observed from the input side of the rotation axis.
(6) FIG. **4** is a cross-sectional view showing a general structure of an actuator including the above harmonic gear device.
(7) FIG. **5**A is a diagrammatic cross-sectional view observed in view of internal and external teeth of the above harmonic gear device.
(8) FIG. **5**B is a cross-sectional view of a line A**1**-A**1** of FIG. **5**A.
(9) FIG. **6** is a conceptual explanation view showing trimming amounts of internal and external teeth of the above harmonic gear device.
(10) FIG. **7**A is a cross-sectional view of a line B**1**-B**1** of FIG. **5**A.
(11) FIG. **7**B is a cross-sectional view of a line B**2**-B**2** of FIG. **5**A.
(12) FIG. **7**C is a cross-sectional view of a line B**3**-B**3** of FIG. **5**A.
(13) FIG. **8**A is a diagrammatic cross-sectional view of periphery of an inner peripheral surface of a flexible external gear of the above harmonic gear device.
(14) FIG. **8**B is an enlarged view of a region Z**1** of FIG. **8**A.
(15) FIG. **9** is a cross-sectional view showing an example of a robot using the above harmonic gear device.
(16) FIG. **10**A is a cross-sectional view of major portions of a harmonic gear device according to a first deformation example of an embodiment.
(17) FIG. **10**B is a cross-sectional view of major portions of a harmonic gear device according to a second deformation example of an embodiment.
(18) FIG. **10**C is a cross-sectional view of major portions of a harmonic gear device according to a third deformation example of an embodiment.

(19) FIG. **10**D is a cross-sectional view of major portions of a harmonic gear device according to a fourth deformation example of an embodiment.
(20) FIG. **11**A is a diagrammatic cross-sectional view observed in view of internal and external teeth of a harmonic gear device according to another embodiment.
(21) FIG. **11**B is a cross-sectional view of a line A**1**-A**1** of FIG. **11**A.
(22) FIG. **12**A is a cross-sectional view of a line B**1**-B**1** of FIG. **11**A.
(23) FIG. **12**B is a cross-sectional view of a line B**2**-B**2** of FIG. **11**A.
(24) FIG. **12**C is a cross-sectional view of a line B**3**-B**3** of FIG. **11**A.
(25) FIG. **13** is a diagrammatic cross-sectional view observed in view of internal and external teeth of a harmonic gear device according to yet another embodiment.
DETAILED DESCRIPTION
(1) Summary
(26) Hereinafter, summary of a harmonic gear device **1** according to an embodiment is explained with reference to FIG. **1**A to FIG. **4**. Drawings referred by embodiments of the disclosure are schematic views, and respective ratios of sizes and thicknesses of structural elements in the drawings do not necessarily reflect actual size ratios. For example, tooth shapes, sizes, tooth numbers, or the like of an internal tooth **21** and an external tooth **31** in FIG. **2** to FIG. **3**B are merely shown schematically for illustration, and are not aimed to be limited to the shapes as shown.
(27) The harmonic gear device **1** according to the embodiment is a gear device including a rigid internal gear **2**, a flexible external gear **3** and a wave generator **4**. In the harmonic gear device **1**, the flexible external gear **3** with an annular shape is arranged on an inner side of the rigid internal gear **2** with an annular shape, and the wave generator **4** is further arranged on an inner side of the flexible external gear **3**. The wave generator **4** deflects the flexible external gear **3** into a non-circular shape, so that external teeth **31** of the flexible external gear **3** are partially meshed with internal teeth **21** of the rigid internal gear **2**. When the wave generator **4** rotates, a meshing position of the internal tooth **21** and the external tooth **31** moves in a circumferential direction of the rigid internal gear **2**, and relative rotation of the flexible external gear **3** according to a difference between tooth numbers of the flexible external gear **3** and the rigid internal gear **2** is generated between the two gears (the rigid internal gear **2** and the flexible external gear **3**). Here, when the rigid internal gear **2** is fixed, the flexible external gear **3** rotates with relative rotation of the two gears. As a result, a rotational output which is decelerated at a relatively high reduction ratio corresponding to the difference between tooth numbers of the two gears, may be obtained from the flexible external gear **3**.
(28) Furthermore, the wave generator **4** deflecting the flexible external gear **3** has a cam **41** having a non-circular shape and driven to rotate with a rotation axis Ax**1** at an input side (referring to FIG. **1**A) as a center, and a bearing **42**. The bearing **42** is arranged between an outer peripheral surface **411** of the cam **41** and an inner peripheral surface **301** of the flexible external gear **3**. An inner ring **422** of the bearing **42** is fixed to the outer peripheral surface **411** of the cam **41**, and an outer ring **421** of the bearing **42** is elastically deformed by being pressed by the cam **41** through ball-shaped rolling bodies **423**. Here the outer ring **421** may relatively rotate relative to the inner ring **422** through rolling of the rolling bodies **423**, so that when the cam **41** with a non-circular shape rotates, rotation of the inner ring **422** is not transmitted to the outer ring **421**, instead, harmonic motion occurs on the external teeth **31** of the flexible external gear **3** pressed by the cam **41**. Since the harmonic motion of the external teeth **31** occurs, the meshing position of the internal tooth **21** and the external tooth **31** moves in the circumferential direction of the rigid internal gear **2** as described above, so that relative rotation occurs between the flexible external gear **3** and the rigid internal gear **2**.
(29) In summary, in such harmonic gear device **1**, the wave generator **4** having the bearing **42** deflects the flexible external gear **3** while transmits power by meshing of the internal teeth **21** and

the external teeth **31**. Therefore, especially when it is used for a long period of time, a foreign matter X**1** (referring to FIG. **8**B) such as metal powder, or nitride, or the like may be generated for example due to absence, or wear, or the like induced by contact between the internal teeth **21** and the external teeth **31**. Due to generation of such foreign matter X**1**, it is possible for the foreign matter X**1** to engage between the internal tooth **21** and the external tooth **31** or for the foreign matter X**1** to enter the bearing **42** of the wave generator **4** to induce damage to the bearing **42**, which may affect reliability of the harmonic gear device **1**. As an example, when the foreign matter X**1** enters the bearing **42**, an indentation generated when the foreign matter X**1** is engaged between the outer ring **421** or the inner ring **422** of the bearing **42** and the rolling body **423** is taken as a starting point, and certain surfaces of the outer ring **421**, the inner ring **422** and the rolling body **423** may be damaged. Such damage (peeling with a type where points occur on surfaces) results in degradation of quality, characteristics, or the like of the harmonic gear device **1**, which results in reduction of reliability of the harmonic gear device **1**. The harmonic gear device **1** of the embodiment is difficult to generate the foreign matter X**1** through the following structures, thereby not easily generating reduction of reliability.

(30) That is, as shown in FIG. **1**A to FIG. **3**B, the harmonic gear device **1** of the embodiment includes: a rigid internal gear **2** with an annular shape, having internal teeth **21**; a flexible external gear **3** with an annular shape, having external teeth **31**; and a wave generator **4**. The flexible external gear **3** is arranged on an inner side of the rigid internal gear **2**. The wave generator **4** is arranged on an inner side of the flexible external gear **3** and deflects the flexible external gear **3**. The harmonic gear device **1** deforms the flexible external gear **3** along with rotation of the wave generator **4** with a rotation axis Ax**1** as a center, such that a part of the external teeth **31** is meshed with a part of the internal teeth **21**, and the flexible external gear **3** relatively rotates relative to the rigid internal gear **2** according to a difference between tooth numbers of the flexible external gear **3** and the rigid internal gear **2**. Here the internal tooth **21** has a tooth-direction trimming portion **210** at an end portion in at least one side along a tooth direction D**1** of the internal tooth **21**.

(31) According to this configuration, the internal tooth **21** has the tooth-direction trimming portion **210** at an end portion in at least one side along the tooth direction D**1** of the internal tooth **21**. An “avoidance portion” is formed between the tooth-direction trimming portion **210** of the internal tooth **21** and the external tooth **31**, therefore, at an end portion in at least one side along the tooth direction D**1** of the internal tooth **21**, it is difficult to generate stress concentration induced by excessive tooth contact between the internal tooth **21** and the external tooth **31** due to the tooth-direction trimming portion **210**. Especially in the harmonic gear device **1**, the wave generator **4** deflects the flexible external gear **3**, therefore deformation of the external tooth **31** such as torsion, deflection (inclination), or the like is sometimes generated with respect to the rotation axis Ax**1**. Therefore, although stress concentration induced by deformation of the external tooth **31** is easily generated at an end portion in at least one side along the tooth direction D**1** of the internal tooth **21**, it is difficult to generate such stress concentration through the tooth-direction trimming portion **210**. Therefore it is difficult to generate the foreign matter X**1** due to absence, or wear, or the like induced by contact between the internal tooth **21** and the external tooth **31**, which may provide the harmonic gear device **1** with high reliability. Furthermore, since the tooth-direction trimming portion **210** is arranged on the rigid internal gear **2**, tooth-direction trimming is unnecessary for the flexible external gear **3**, or a trimming amount of the flexible external gear **3** may be reduced, and reduction of strength of the flexible external gear **3** induced by performing tooth-direction trimming on the flexible external gear **3** with flexibility is easily suppressed.

(32) Furthermore, in the harmonic gear device **1** of some embodiments, surface hardness of the internal tooth **21** is lower than that of the external tooth **31**. The external tooth **31** protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21**.

(33) According to this configuration, the external tooth **31** with relatively high surface hardness protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21**,

therefore, in at least one side along the tooth direction D**1**, it is difficult to generate a height difference induced by wear at a tooth surface of the internal tooth **21**. That is, in at least one side along the tooth direction D**1**, the internal tooth **21** with relatively low surface hardness is evenly worn due to tooth contact between it and the external tooth **31**, so that it is difficult to generate a local recess (height difference) at the tooth surface of the internal tooth **21**. Therefore, even though there is a situation where a tooth contact position deviates in the tooth direction D**1** due to certain jitter, occurrence of abnormality of the harmonic gear device **1** due to an excessive load acting at the meshing position of the internal tooth **21** and the external tooth **31** is easily suppressed. That is, it is difficult to generate absence of a corner portion such as an end portion along a tooth direction D**1** of the external tooth **31**, and as a result, it is difficult to generate a hard foreign matter X**1** (with a relatively high hardness). Therefore it is difficult to generate the foreign matter X**1** due to absence, or wear, or the like induced by contact between the internal tooth **21** and the external tooth **31**, and the harmonic gear device **1** with high reliability may be provided.
(34) In summary, according to the harmonic gear device **1** of some embodiments, since it is difficult to generate the foreign matter X**1**, an effect of high reliability may be achieved. Furthermore, it is difficult to reduce reliability of the harmonic gear device **1** of some embodiments, especially even when it is used for a long period of time, thereby achieving a long service life and high performance of the harmonic gear device **1**.
(35) Furthermore, as shown in FIG. **4**, the harmonic gear device **1** of some embodiments constitutes an actuator **100** together with a driving source **101** and an output portion **102**. In other words, the actuator **100** of some embodiments includes the harmonic gear device **1**, the driving source **101** and the output portion **102**. The driving source **101** is configured to rotate the wave generator **4**. The output portion **102** is configured to take a rotation force of one of the rigid internal gear **2** or the flexible external gear **3** out as output.
(36) The actuator **100** according to some embodiments has an advantage of being difficult to reduce reliability of the harmonic gear device **1**.
(2) Definition
(37) “annular shape” stated in some embodiments of the disclosure refers to a shape such as a ring (circle) which forms an enclosed space (area) on an inner side when it is observed at least in a top view, and is not limited to a perfect circle or a circular shape (circular ring shape) when it is observed in a top view, for example, may also be an elliptical shape, a polygonal shape, or the like. Furthermore, the annular shape may also be a shape such as a cup-shaped flexible external gear **3** with a bottom portion **322**, and the flexible external gear is referred to as an “annular” flexible external gear **3** when a trunk portion **321** thereof has an annular shape.
(38) “tooth-direction trimming” stated in some embodiments of the disclosure refers to trimming in the tooth direction D**1**, and the tooth-direction trimming portion **210** of the internal tooth **21** is a portion of the internal tooth **21** where tooth-direction trimming is performed. By tooth-direction trimming, a regular tooth-direction shape of a gear may be provided with a conscious bulge or change its torsion angle. As representative processes of tooth-direction trimming, there are drum-shaped trimming and edge trimming process (tooth end trimming). The drum-shaped trimming is a process by which a central portion along a tooth direction D**1** of a gear is convex so that a central portion toward the tooth direction D**1** has rounded corners. The edge trimming process is a processing method for appropriately avoiding both end portions in the tooth direction D**1**. The drum-shaped trimming is a process performed substantially throughout the entire length in the tooth direction D**1** with rounded corners facing toward the central portion. In comparison, the edge trimming process is a process avoiding both end portions in the tooth direction D**1** only. No matter the drum-shaped trimming or the edge trimming process, tooth thickness of each of both end portions in the tooth direction D**1** is less than that of the central portion, so that a tooth contact position with a gear at an opposite side may be close to center in the tooth direction D**1**. By such tooth-direction trimming, it is possible to suppress “an end contact” in which tooth contact is close

to an end portion in the tooth direction D**1**, due to a manufacturing error or assembly error of the gear, especially stress concentration of the end portion in the tooth direction D**1** (a tooth width end) is mitigated, and tooth contact is improved.
(39) "foreign matter" stated in some embodiments of the disclosure refers to a substance other than original constituent elements of the harmonic gear device **1**, and as an example, there is metal powder, or nitride, or the like generated due to absence, or wear, or the like induced by contact between the internal tooth **21** and the external tooth **31**. Furthermore, the foreign matter X**1** which is hindered from entering by a lubricant Lb**1** (referring to FIG. **8**B) as described below or the like, is not limited to a substance generated inside the harmonic gear device **1**, for example, it includes waste, dust, dirt, or the like entering from outside of the harmonic gear device **1**. "hinder" as stated here refers to interference or obstruction, and is not limited to complete prevention, and includes all situations where the foreign matter X**1** is difficult to enter.
(40) "it is difficult to generate the foreign matter X**1**" stated in some embodiments of the disclosure refers to reduction of at least one of generation amount, generation rate or generation frequency of the foreign matter X**1** (hereinafter, referred to as "generation amount or the like"). Here the generation amount or the like is a value of the foreign matter X**1** especially related to hardness and size which become causes of reduction of reliability of the harmonic gear device **1**, and a situation where it is difficult to generate the foreign matter X**1**, includes for example a situation where generation amount or the like of a hard foreign matter X**1** (with a relatively high hardness) is reduced. Of course, a situation where the foreign matter X**1** is not generated completely, is also included in the situation where it is difficult to generate the foreign matter X**1**. That is, in some embodiments, generation amount or the like of the foreign matter X**1** is reduced by a structure where the tooth-direction trimming portion **210** is arranged at an end portion in at least one side along the tooth direction D**1** of the internal tooth **21**, compared with a situation where the structure is not used. Similarly, in some embodiments, generation amount or the like of the foreign matter X**1** is reduced by a structure where the external tooth **31** with relatively high surface hardness protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21**, compared with a situation where the structure is not used.
(41) "rigid" stated in some embodiments of the disclosure refers to a property of an object resisting deformation when an external force is applied to the object and the object is to be deformed. In other words, a rigid object is difficult to be deformed even when an external force is applied thereto. Furthermore, "flexible" stated in the embodiments of the disclosure refers to a property of elastic deformation (deflection) of an object when an external force is applied to the object. In other words, a flexible object is easy to be elastically deformed when an external force is applied thereto. Therefore, "rigid" and "flexible" have opposite meanings.
(42) Especially in some embodiments of the disclosure, "rigid" of the rigid internal gear **2** and "flexible" of the flexible external gear **3** are used in opposite meanings. That is, "rigid" of the rigid internal gear **2** refers to that the rigid internal gear **2** has a relatively high rigidity, compared with at least the flexible external gear **3**, that is, the rigid internal gear **2** is difficult to be deformed even when an external force is applied thereto. Similarly, "flexible" of the flexible external gear **3** refers to that the flexible external gear **3** has a relatively high flexibility, compared with at least the rigid internal gear **2**, that is, the flexible external gear **3** is easy to be elastically deformed when an external force is applied thereto.
(43) Furthermore, in some embodiments of the disclosure, sometimes one side (right side of FIG. **1**A) of the rotation axis Ax**1** is referred to as an "input side", and the other side (left side of FIG. **1**A) of the rotation axis Ax**1** is referred to as an "output side". That is, in the example of FIG. **1**A, the flexible external gear **3** has an opened surface **35** at "input side" of the rotation axis Ax**1**. However, "input side" and "output side" are merely labels added for illustration, and are not aimed to limit a positional relationship between input and output observed from the harmonic gear device **1**.

(44) “non-circular shape” in some embodiments of the disclosure refers to a shape without a perfect circle, including for example an elliptical shape, an oblong shape, or the like. In some embodiments, as an example, the cam **41** with a non-circular shape of the wave generator **4** has an elliptical shape. That is, in some embodiments, the wave generator **4** deflects the flexible external gear **3** into an elliptical shape.
(45) “elliptical shape” stated in some embodiments of the disclosure refers to all shapes where a perfect circle is pressed so that an intersection point of a long axis and a short axis orthogonal to each other is located at the center, and is not limited to a curve formed by a set of points where a sum of distances of the points from two fixed points on a plane is constant, i.e., a mathematical “ellipse”. That is, the cam **41** of the embodiment may be a curve formed by a set of points where a sum of distances of the points from two fixed points on a plane is constant, such as a mathematical “ellipse”, or may not be a mathematical “ellipse”, instead, it has an elliptical shape like an oblong shape. As described above, drawings referred by some embodiments of the disclosure are schematic views, and respective ratios of sizes and thicknesses of structural elements in the drawings do not necessarily reflect actual size ratios. Therefore, for example in FIG. **2**, shape of the cam **41** of the wave generator **4** is taken as a slightly exaggerated elliptical shape, however, it is not aimed to limit an actual shape of the cam **41**.
(46) “rotation axis” stated in some embodiments of the disclosure refers to a virtual axis (straight line) which is center of a rotational motion of a rotation body. That is, the rotation axis Ax**1** is a virtual axis without an entity. The wave generator **4** rotates with the rotation axis Ax**1** as the center.
(47) Each of “internal teeth” and “external teeth” stated in some embodiments of the disclosure refers to a set (group) of multiple “teeth” rather than a “tooth” as a single body. That is, the internal teeth **21** of the rigid internal gear **2** include a set of teeth formed on an inner peripheral surface of the rigid internal gear **2**. Similarly, the external teeth **31** of the flexible external gear **3** include a set of teeth formed on an outer peripheral surface of the flexible external gear **3**.
(48) Furthermore, “parallel” stated in some embodiments of the disclosure includes a situation where when two straight lines are on a plane, the two straight lines do not intersect with each other no matter where they extend, that is, an angle between the two straight lines is strictly 0 degrees (or 180 degrees), besides an error range relationship where the angle between the two straight lines is at a relative 0 degrees with an extent of converging to a few degrees (for example, less than 10 degrees). Similarly, “orthogonal” stated in some embodiments of the disclosure includes a situation where the two straight lines intersect with each other strictly by an angle of 90 degrees there-between, besides an error range relationship where the angle between the two straight lines is at a relative 90 degrees with an extent of converging to a few degrees (for example, less than 10 degrees).
(3) Structure
(49) Hereinafter, detailed structures of the harmonic gear device **1** and the actuator **100** of the embodiment are described with reference to FIG. **1**A to FIG. **6**C.
(50) FIG. **1**A is a cross-sectional view showing a general structure of a harmonic gear device **1**. FIG. **1**B is an enlarged view of a region Z**1** of FIG. **1**A. FIG. **2** is a diagrammatic view of the harmonic gear device **1** observed from an input side of a rotation axis Ax**1** (right side of FIG. **1**A). FIG. **3**A is a diagrammatic exploded perspective view of the harmonic gear device **1** observed from an output side of the rotation axis Ax**1** (left side of FIG. **1**A). FIG. **3**B is a diagrammatic exploded perspective view of the harmonic gear device **1** observed from the input side of the rotation axis Ax**1**. FIG. **4** is a cross-sectional view showing a general structure of an actuator **100** including the harmonic gear device **1**.
(51) (3.1) Harmonic Gear Device
(52) As described above, the harmonic gear device **1** of the embodiment includes a rigid internal gear **2**, a flexible external gear **3** and a wave generator **4**. In some embodiments, the rigid internal gear **2**, the flexible external gear **3** and the wave generator **4** as structural elements of the harmonic

gear device **1** are made of metal materials such as stainless steel, cast iron, carbon steel for mechanical structure, chrome molybdenum steel, phosphor bronze, aluminum bronze, or the like. Metal as stated here includes a metal which is subject to surface treatments such as nitridation or the like.
(53) Furthermore, in some embodiments, a cup-shaped harmonic gear device is exemplified as an example of the harmonic gear device **1**. That is, in the harmonic gear device **1** of some embodiments, the flexible external gear **3** formed into a cup shape is used. The wave generator **4** is combined with the flexible external gear **3** in a manner of being received in the cup-shaped flexible external gear **3**.
(54) Furthermore, in some embodiments, as an example, the harmonic gear device **1** is used in a state of fixing the rigid internal gear **2** to an input-side housing **111** (referring to FIG. **4**) and an output-side housing **112** (referring to FIG. **4**) or the like. Therefore, the flexible external gear **3** relatively rotates relative to fixed components (the input-side housing **111**, or the like) along with relative rotation of the rigid internal gear **2** and the flexible external gear **3**.
(55) Furthermore, in some embodiments, when the harmonic gear device **1** is used in the actuator **100**, a rotation force as output is taken out from the flexible external gear **3** by applying a rotation force as input to the wave generator **4**. That is, the harmonic gear device **1** acts with rotation of the wave generator **4** as input rotation and with rotation of the flexible external gear **3** as output rotation. Therefore, in the harmonic gear device **1**, output rotation which is decelerated at a relatively high reduction ratio with respect to input rotation may be obtained.
(56) Furthermore, in the harmonic gear device **1** of some embodiments, the rotation axis Ax**1** on the input side and a rotation axis Ax**2** on the output side are on the same line. In other words, the rotation axis Ax**1** on the input side is coaxial to the rotation axis Ax**2** on the output side. Here the rotation axis Ax**1** on the input side is center of rotation of the wave generator **4** to which input rotation is applied, and the rotation axis Ax**1** on the output side is center of rotation of the flexible external gear **3** for generating output rotation. That is, in the harmonic gear device **1**, output rotation which is decelerated at a relatively high reduction ratio with respect to input rotation may be obtained on the same axis.
(57) The rigid internal gear **2** is also referred to as a circular spline which is an annular component having internal teeth **21**. In some embodiments, the rigid internal gear **2** has a circular ring shape of which at least an inner peripheral surface is a perfect circle when it is observed in a top view. In the inner peripheral surface of the rigid internal gear **2** with the circular ring shape, the internal teeth **21** are formed along the circumferential direction of the rigid internal gear **2**. Multiple teeth constituting the internal teeth **21** are all in the same shape, and are arranged at equal intervals in the whole area along the circumferential direction of the inner peripheral surface of the rigid internal gear **2**. That is, a pitch circle of the internal tooth **21** is a perfect circle when it is observed in a top view. Furthermore, the rigid internal gear **2** has a predetermined thickness in a direction of the rotation axis Ax**1**. The internal teeth **21** are all formed throughout the entire length in a thickness direction of the rigid internal gear **2**. Tooth lines of the internal teeth **21** are all parallel to the rotation axis Ax**1**.
(58) As described above, the rigid internal gear **2** is fixed to the input-side housing **111** (referring to FIG. **4**) and the output-side housing **112** (referring to FIG. **4**) or the like. Therefore, multiple fixing holes **22** for fixing (referring to FIG. **3**A and FIG. **3**B) are formed in the rigid internal gear **2**.
(59) The flexible external gear **3** is also referred to as a flex spline which is an annular component having external teeth **31**. In some embodiments, the flexible external gear **3** is formed as a cup-shaped component by a thin-wall metal elastomer (metal plate). That is, the flexible external gear **3** has flexibility due to its small (thin) thickness. The flexible external gear **3** has a cup-shaped body portion **32**. The body portion **32** has a trunk portion **321** and a bottom portion **322**. In a state where the flexible external gear **3** is not elastically deformed, at least an inner peripheral surface **301** of the trunk portion **321** has a cylindrical shape which is a perfect circle when it is observed in a top

view. A central axis of the trunk portion **321** is consistent with the rotation axis Ax**1**. The bottom portion **322** is arranged on an opened surface at a side of the trunk portion **321** and has a disc shape which is a perfect circle when it is observed in a top view. The bottom portion **322** is arranged on an opened surface, close to the output side of the rotation axis Ax**1**, of a pair of opened surfaces of the trunk portion **321**. According to the above contents, the body portion **32** achieves, through a whole of the trunk portion **321** and the bottom portion **322**, a cylindrical shape with bottom, i.e., a cup shape which is opened toward the input side of the rotation axis Ax**1**. In other words, an opened surface **35** is formed on an end surface at a side of the flexible external gear **3** opposite to the bottom portion **322** in the direction of the rotation axis Ax**1**. That is, the flexible external gear **3** has a cylindrical shape with an opened surface **35** at a side along the tooth direction D**1** (here the input side of the rotation axis Ax**1**). In some embodiments, the trunk portion **321** and the bottom portion **322** are integrally formed by a metal member, thereby obtaining a seamless body portion **32**.

(60) Here the wave generator **4** with a non-circular shape (elliptical shape) is embedded in an inner side of the trunk portion **321**, so that the wave generator **4** is combined with the flexible external gear **3**. Therefore, the flexible external gear **3** bears, from the wave generator **4**, an external force in a radial direction (a direction orthogonal to the rotation axis Ax**1**) from the inner side to the outer side, so that the flexible external gear **3** is elastically deformed into a non-circular shape. In some embodiments, the wave generator **4** is combined with the flexible external gear **3**, so that the trunk portion **321** of the flexible external gear **3** is elastically deformed into an elliptical shape. That is, a state where the flexible external gear **3** is not elastically deformed refers to a state where the flexible external gear **3** is not combined with the wave generator **4**. In contrast, a state where the flexible external gear **3** is elastically deformed refers to a state where the flexible external gear **3** is combined with the wave generator **4**.

(61) More specifically, the wave generator **4** is embedded in an end portion at a side of the inner peripheral surface **301** of the trunk portion **321** opposite to the bottom portion **322** (at the input side of the rotation axis Ax**1**). In other words, the wave generator **4** is embedded in an end portion at the opened surface **35** side of the trunk portion **321** of the flexible external gear **3** in the direction of the rotation axis Ax**1**. Therefore, in a state where the flexible external gear **3** is elastically deformed, an end portion at the opened surface **35** side of the flexible external gear **3** in the direction of the rotation axis Ax**1** is deformed greater than deformation of an end portion at the bottom portion **322** side, to form a shape closer to an elliptical shape. Due to such difference between deformation amounts in the direction of the rotation axis Ax**1**, the inner peripheral surface **301** of the trunk portion **321** of the flexible external gear **3** includes a tapered surface **302** (referring to FIG. **8**A) inclined with respect to the rotation axis Ax**1**, in a state where the flexible external gear **3** is elastically deformed.

(62) Furthermore, at least an end portion at a side of an outer peripheral surface of the trunk portion **321** opposite to the bottom portion **322** (at the input side of the rotation axis Ax**1**) is formed with external teeth **31** in a circumferential direction of the trunk portion **321**. In other words, the external teeth **31** are arranged at at least an end portion in the opened surface **35** side of the trunk portion **321** of the flexible external gear **3** in the direction of the rotation axis Ax**1**. Multiple teeth constituting the external teeth **31** are all in the same shape, and are arranged at equal intervals in the whole area along a circumferential direction of the outer peripheral surface of the flexible external gear **3**. That is, in a state where the flexible external gear **3** is not elastically deformed, a pitch circle of the external tooth **31** is a perfect circle when it is observed in a top view. The external teeth **31** are formed only in a range of a certain width of the trunk portion **321** from an end edge at the opened surface **35** side (the input side of the rotation axis AX**1**). Specifically, in at least a portion (an end portion at the opened surface **35** side), in which the wave generator **4** embedded, of the trunk portion **321** in the direction of the rotation axis Ax**1**, the external teeth **31** are formed on an outer peripheral surface thereof. Tooth directions of the external teeth **31** are all parallel to the

rotation axis Ax**1**.
(63) In summary, in the harmonic gear device **1** of some embodiments, tooth directions of the internal teeth **21** of the rigid internal gear **2** and tooth directions of the external teeth **31** of the flexible external gear **3** are all parallel to the rotation axis Ax**1**. Therefore, in some embodiments, “tooth direction D**1**” is a direction parallel to the rotation axis Ax**1**. Furthermore, size of the internal tooth **21** in the tooth direction D**1** is a tooth width of the internal tooth **21**, and similarly, size of the external tooth **31** in the tooth direction D**1** is a tooth width of the external tooth **31**, so that the tooth direction D**1** has the same meaning as a tooth width direction.
(64) In some embodiments, as described above, rotation of the flexible external gear **3** is taken out as output rotation. Therefore an output portion **102** (referring to FIG. **4**) of the actuator **100** is mounted on the flexible external gear **3**. The bottom portion **322** of the flexible external gear **3** is formed with multiple mounting holes **33** for mounting a shaft used as the output portion **102**. Furthermore, a through-hole **34** is formed in a central portion of the bottom portion **322**. A wall around the through-hole **34** of the bottom portion **322** is thicker than walls at other portions of the bottom portion **322**
(65) The flexible external gear **3** configured as such is arranged on the inner side of the rigid internal gear **2**. Here the flexible external gear **3** is combined with the rigid internal gear **2** in a manner that only an end portion at a side of the outer peripheral surface of the trunk portion **321** opposite to the bottom portion **322** (at the input side of the rotation axis Ax**1**) is inserted into the inner side of the rigid internal gear **2**. That is, the flexible external gear **3** inserts a portion (an end portion at the opened surface **35** side), in which the wave generator **4** embedded, of the trunk portion **321** in the direction of the rotation axis Ax**1**, into the inner side of the rigid internal gear **2**. Here the external teeth **31** are formed on the outer peripheral surface of the flexible external gear **3**, and the internal teeth **21** are formed on the inner peripheral surface of the rigid internal gear **2**. Therefore, in case that the flexible external gear **3** is arranged on the inner side of the rigid internal gear **2**, the external teeth **31** and the internal teeth **21** are opposite to each other.
(66) Here the tooth number of the internal teeth **21** of the rigid internal gear **2** is greater than that of the external teeth **31** of the flexible external gear **3** by 2N (N is a positive integer). As an example of some embodiments, N is “1”, and the tooth number (of the external teeth **31**) of the flexible external gear **3** is greater than that (of the internal teeth **21**) of the rigid internal gear **2** by “2”. Such difference between tooth numbers of the flexible external gear **3** and the rigid internal gear **2** specifies a reduction ratio of output rotation with respect to input rotation in the harmonic gear device **1**.
(67) Here, in some embodiments, as an example, as shown in FIG. **1**A and FIG. **1**, a relative position of the flexible external gear **3** and the rigid internal gear **2** in the direction of the rotation axis Ax**1** is set in a manner that center of the external tooth **31** in the tooth direction D**1** is opposite to center of the internal tooth **21** in the tooth direction D**1**. That is, as to the external teeth **31** of the flexible external gear **3** and the internal teeth **21** of the rigid internal gear **2**, central positions thereof in the tooth direction D**1** is aligned with the same position in the direction of the rotation axis Ax**1**. Furthermore, in some embodiments, the size (tooth width) of the external tooth **31** in the tooth direction D**1** is greater than the size (tooth width) of the internal tooth **21** in the tooth direction D**1**. Therefore, in the direction parallel to the rotation axis Ax**1**, the internal tooth **21** is received in a range of the external tooth **31** in the tooth direction. In other words, the external tooth **31** protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21**. Details are described in a “(4.3) Tooth Width” column, and in some embodiments, the external tooth **31** protrudes toward both sides along the tooth direction D**1** (input and output sides of the rotation axis Ax**1**) relative to the internal tooth **21**.
(68) Here, in a state where the flexible external gear **3** is not elastically deformed (in a state where the flexible external gear **3** is not combined with the wave generator **4**), a pitch circle of the external tooth **31** depicted as a perfect circle is set to be smaller than a pitch circle of the internal

tooth **21** depicted as a perfect circle too. That is, in a state where the flexible external gear **3** is not elastically deformed, the external tooth **31** and the internal tooth **21** are opposite to each other with a gap there-between and are not meshed with each other.

(69) On the other hand, in a state where the flexible external gear **3** is elastically deformed (in a state where the flexible external gear **3** is combined with the wave generator **4**), the trunk portion **321** is deflected into an elliptical shape (non-circular shape), so that the external teeth **31** of the flexible external gear **3** are partially meshed with the internal teeth **21** of the rigid internal gear **2**. That is, (at least an end portion at the opened surface **35** side of) the trunk portion **321** of the flexible external gear **3** is elastically deformed into an elliptical shape, therefore, the external teeth **31** at both end portions in a long-axis direction of the elliptical shape are meshed with the internal teeth **21**, as shown in FIG. **2**. In other words, a long diameter of a pitch circle of the external tooth **31** depicted as an ellipse is consistent with a diameter of a pitch circle of the internal tooth **21** depicted as a perfect circle, and a short diameter of the pitch circle of the external tooth **31** depicted as an ellipse is less than the diameter of the pitch circle of the internal tooth **21** depicted as a perfect circle. In this way, when the flexible external gear **3** is elastically deformed, a part of teeth constituting the external teeth **31** is meshed with a part of teeth constituting the internal teeth **21**. As a result, in the harmonic gear device **1**, a part of the external teeth **31** may be meshed with a part of the internal teeth **21**.

(70) The wave generator **4** is also referred to as a wave form generator which is a component configured to deflect the flexible external gear **3** so that harmonic motion occurs on the external teeth **31** of the flexible external gear **3**. In some embodiments, the wave generator **4** is a component of which an outer peripheral shape is a non-circular shape, especially an elliptical shape when it is observed in a top view.

(71) The wave generator **4** has a cam **41** with a non-circular shape (here an elliptical shape) and a bearing **42** assembled on periphery of the cam **41**. That is, the cam **41** is combined with the bearing **42** by embedding the cam **41** with a non-circular shape (elliptical shape) in an inner side of an inner ring **422** of the bearing **42**. Therefore, the bearing **42** bears, from the cam **41**, an external force in a radial direction (a direction orthogonal to the rotation axis Ax**1**) from the inner side to an outer side of the inner ring **422**, so that the bearing **42** is elastically deformed into a non-circular shape. That is, a state where the bearing **42** is not elastically deformed refers to a state where the bearing **42** is not combined with the cam **41**. In contrast, a state where the bearing **42** is elastically deformed refers to a state where the bearing **42** is combined with the cam **41**.

(72) The cam **41** is a component with a non-circular shape (here an elliptical shape) which is driven to rotate with a rotation axis Ax**1** at an input side as a center. The cam **41** has an outer peripheral surface **411** (referring to FIG. **1**), and at least the outer peripheral surface **411** is composed of a metal plate with an elliptical shape when it is observed in a top view. The cam **41** has a predetermined thickness in the direction of the rotation axis Ax**1** (i.e., the tooth direction D**1**). Therefore, the cam **41** has rigidity equal to that of the rigid internal gear **2**. However, thickness of the cam **41** is less (thinner) than that of the rigid internal gear **2**. In some embodiments, as described above, rotation of the wave generator **4** is taken as input rotation. Therefore, an input portion **103** (referring to FIG. **4**) of the actuator **100** is mounted on the wave generator **4**. A central portion of the cam **41** of the wave generator **4** is formed with a cam hole **43** for mounting a shaft used as the input portion **103**.

(73) The bearing **42** has an outer ring **421**, an inner ring **422**, and multiple rolling bodies **423**. In some embodiments, as an example, the bearing **42** includes a deep groove ball bearing using balls as the rolling bodies **423**.

(74) Each of the outer ring **421** and the inner ring **422** is an annular component. Each of the outer ring **421** and the inner ring **422** is formed as an annular component by a thin-wall metal elastomer (metal plate). That is, each of the outer ring **421** and the inner ring **422** has flexibility due to its small (thin) thickness. In some embodiments, each of the outer ring **421** and the inner ring **422** has

a circular ring shape which is a perfect circle when it is observed in a top view, in a state where the bearing **42** is not elastically deformed (in a state where the cam **41** is combined with the bearing **42**). The inner ring **422** is smaller than the outer ring **421** and is arranged on an inner side of the outer ring **421**. Here an inner diameter of the outer ring **421** is greater than an outer diameter of the inner ring **422**, and thus a gap is generated between an inner peripheral surface of the outer ring **421** and an outer peripheral surface of the inner ring **422**.
(75) Multiple rolling bodies **423** are arranged in the gap between the outer ring **421** and the inner ring **422**. Multiple rolling bodies **423** are arranged along a circumferential direction of the outer ring **421**. Multiple rolling bodies **423** are all metal balls (balls) with the same shape, and are arranged at equal intervals in the whole area along the circumferential direction of the outer ring **421**. Here, although not specifically illustrated, the bearing **42** also has a retainer, and multiple rolling bodies **423** are held between the outer ring **421** and the inner ring **422** by the retainer.
(76) Furthermore, in some embodiments, as an example, size of each of the outer ring **421** and the inner ring **422** in a width direction (a direction parallel to the rotation axis Ax**1**) is the same as the thickness of the cam **41**. That is, size of each of the outer ring **421** and the inner ring **422** in the width direction is less than the thickness of the rigid internal gear **2**.
(77) With such structure of the bearing **42**, the cam **41** is combined with the bearing **42**, so that the inner ring **422** of the bearing **42** is fixed to the cam **41**, and the inner ring **422** is elastically deformed into an elliptical shape similar to a peripheral shape of the cam **41**. At this time, the outer ring **421** of the bearing **42** is pressed by the inner ring **422** via multiple rolling bodies **423**, to be elastically deformed into an elliptical shape. Therefore each of the outer ring **421** and the inner ring **422** of the bearing **42** is elastically deformed into an elliptical shape. In this way, the outer ring **421** and the inner ring **422** have elliptical shapes similar to each other respectively, in a state where the bearing **42** is elastically deformed (in a state where the cam **41** is combined with the bearing **42**).
(78) Even in a state where the bearing **42** is elastically deformed, since multiple rolling bodies **423** are interposed between the outer ring **421** and the inner ring **422**, the gap between the outer ring **421** and the inner ring **422** is maintained to be substantially constant throughout the entire circumference of the outer ring **421**. Furthermore, in this state, the outer ring **421** may relatively rotate relative to the inner ring **422** by rolling multiple rolling bodies **423** between the outer ring **421** and the inner ring **422**. Therefore, in a state where the bearing **42** is elastically deformed, when the cam **41** rotates with the rotation axis Ax**1** as the center, rotation of the cam **41** is not transmitted to the outer ring **421**, and elastic deformation of the inner ring **422** is transmitted to the outer ring **421** through multiple rolling bodies **423**. That is, in the wave generator **4**, when the cam **41** rotates with the rotation axis Ax**1** as the center, the outer ring **421** is elastically deformed in a manner that a long axis of an elliptical shape imitated by the outer ring **421** rotates with the rotation axis Ax**1** as the center. Therefore, as to the whole wave generator **4**, an outer peripheral shape of the wave generator **4** which has an elliptical shape when it is observed from the input side of the rotation axis Ax**1**, is changed along with rotation of the cam **41** in a manner that a long axis of the elliptical shape rotates with the rotation axis Ax**1** as the center.
(79) The wave generator **4** configured as such is arranged on the inner side of the flexible external gear **3**. Here the flexible external gear **3** is combined with the wave generator **4** in a manner that only an end portion at a side of the inner peripheral surface **301** of the trunk portion **321** opposite to the bottom portion **322** (at the opened surface **35** side) is embedded in the wave generator **4**. At this time, the bearing **42** of the wave generator **4** is arranged between the outer peripheral surface **411** of the cam **41** and the inner peripheral surface **301** of the flexible external gear **3**. Here outer diameter of the outer ring **421** in a state where the bearing **42** is not elastically deformed (in a state where the cam **41** is not combined with the bearing **42**) is the same as inner diameter of the flexible external gear **3** (the trunk portion **321**) in a state where it is not elastically deformed either. Therefore an outer peripheral surface of the outer ring **421** of the wave generator **4** is connected to the inner peripheral surface **301** of the flexible external gear **3** throughout the entire circumference in a

circumferential direction of the bearing **42**. Therefore, in a state where the flexible external gear **3** is elastically deformed (in a state where the wave generator **4** is combined with the flexible external gear **3**), the trunk portion **321** is deflected into an elliptical shape (non-circular shape). In this state, the flexible external gear **3** is fixed relative to the outer ring **421** of the bearing **42**.

(80) In the harmonic gear device **1** with the above structure, as shown in FIG. **2**, the trunk portion **321** of the flexible external gear **3** is deflected into an elliptical shape (non-circular shape), so that the external teeth **31** of the flexible external gear **3** are partially meshed with the internal teeth **21** of the rigid internal gear **2**. That is, (the trunk portion **321** of) the flexible external gear **3** is elastically deformed into an elliptical shape, so that the external teeth **31** at two positions corresponding to both end portions in a long-axis direction of the elliptical shape are meshed with the internal teeth **21**. Furthermore, when the cam **41** rotates with the rotation axis Ax**1** as the center, rotation of the cam **41** is not transmitted to the outer ring **421** and the flexible external gear **3**, and elastic deformation of the inner ring **422** is transmitted to the outer ring **421** and the flexible external gear **3** through multiple rolling bodies **423**. Therefore, an outer peripheral shape of the flexible external gear **3** which has an elliptical shape when it is observed from the input side of the rotation axis Ax**1**, is changed along with rotation of the cam **41** in a manner that a long axis of the elliptical shape rotates with the rotation axis Ax**1** as the center.

(81) As a result, fluctuation motion occurs on the external teeth **31** formed on the outer peripheral surface of the flexible external gear **3**. Due to occurrence of fluctuation motion of the external teeth **31**, a meshing position of the internal tooth **21** and the external tooth **31** moves in a circumferential direction of the rigid internal gear **2**, so that relative rotation occurs between the flexible external gear **3** and the rigid internal gear **2**. That is, the external teeth **31** are meshed with the internal teeth **21** at both end portions in a long-axis direction of an elliptical shape formed by (the trunk portion **321** of) the flexible external gear **3**, therefore, the meshing position of the internal tooth **21** and the external tooth **31** moves by rotating a long axis of the elliptical shape with the rotation axis Ax**1** as the center. In this way, as to the harmonic gear device **1** according to some embodiments, the flexible external gear **3** is deformed along with rotation of the wave generator **4** with the rotation axis Ax**1** as the center, so that a part of the external teeth **31** is meshed with a part of the internal teeth **21**, thereby rotating the flexible external gear **3** according to a difference between tooth numbers of the flexible external gear **3** and the rigid internal gear **2**.

(82) However, in the harmonic gear device **1**, as described above, the difference between tooth numbers of the flexible external gear **3** and the rigid internal gear **2** specifies a reduction ratio of output rotation with respect to input rotation in the harmonic gear device **1**. That is, in case that the tooth number of the rigid internal gear **2** is set to “V**1**” and the tooth number of the flexible external gear **3** is set to “V**2**”, a reduction ratio R**1** is represented by Formula 1 below.

$R1=V2/(V1-V2)$ (Formula 1)

(83) In summary, the smaller the difference (V**1**−V**2**) between tooth numbers of the rigid internal gear **2** and the flexible external gear **3**, the larger the reduction ratio R**1**. As an example, the tooth number V**1** of the rigid internal gear **2** is “72”, the tooth number V**2** of the flexible external gear **3** is “70”, and the difference (V**1**−V**2**) between tooth numbers thereof is “2”, and thus the reduction ratio R**1** is “35” according to the above Formula 1. In this case, when it is observed from the input side of the rotation axis Ax**1**, the cam **41** rotates clockwise by a circle (360 degrees) with the rotation axis Ax**1** as the center, then the flexible external gear **3** rotates counterclockwise by an amount equal to the difference “2” between tooth numbers (i.e., 10.3 degrees) with the rotation axis Ax**1** as the center.

(84) According to the harmonic gear device **1** of some embodiments, such high reduction ratio R**1** may be achieved by a combination of primary gears (the rigid internal gear **2** and the flexible external gear **3**).

(85) Furthermore, it is feasible as long as the harmonic gear device **1** includes at least the rigid internal gear **2**, the flexible external gear **3** and the wave generator **4**. For example, the harmonic

gear device **1** may further include a spline bushing **113** as described in a "(3.2) Actuator" column or the like as structural elements.
(86) (3.2) Actuator
(87) Next, a structure of the actuator **100** of the embodiment is described in more detail.
(88) As shown in FIG. **4**, the actuator **100** of the embodiment includes the harmonic gear device **1** of some embodiments, a driving source **101** and an output portion **102**. That is, the actuator **100** includes the driving source **101** and the output portion **102**, besides the rigid internal gear **2**, the flexible external gear **3** and the wave generator **4** constituting the harmonic gear device **1**. Furthermore, the actuator **100** includes an input portion **103**, an input-side housing **111**, an output-side housing **112**, a spline bushing **113**, a spacer **114**, a first stopper **115**, a second stopper **116** and a mounting plate **117**, besides the harmonic gear device **1**, the driving source **101** and the output portion **102**. Furthermore, in some embodiments, the actuator **100** further includes input-side bearings **118**, **119**, an input-side oil seal **120**, output-side bearings **121**, **122**, and an output-side oil seal **123**.
(89) In some embodiments, components of the actuator **100** except the driving source **101**, the input-side oil seal **120** and the output-side oil seal **123** are made of metal materials such as stainless steel, cast iron, carbon steel for mechanical structure, chrome molybdenum steel, phosphor bronze, aluminum bronze, or the like.
(90) The driving source **101** is a power generation source such as an electric machine (motor) or the like. Power generated by the driving source **101** is transmitted to the cam **41** of the wave generator **4** in the harmonic gear device **1**. Specifically, the driving source **101** is connected to a shaft used as the input portion **103**, and the power generated by the driving source **101** is transmitted to the cam **41** via the input portion **103**. Therefore, the driving source **101** may rotate the cam **41**.
(91) The output portion **102** is a cylindrical shaft arranged along the rotation axis Ax**2** on the output side. A central axis of the shaft used as the output portion **102** is consistent with the rotation axis Ax**2**. The output portion **102** is held by the output-side housing **112** to be rotatable with the rotation axis Ax**2** as a center. The output portion **102** is fixed to the bottom portion **322** of the body portion **32** in the flexible external gear **3** and rotates with the rotation axis Ax**2** as the center, together with the flexible external gear **3**. That is, the output portion **102** takes a rotation force of the flexible external gear **3** out as output.
(92) The input portion **103** is a cylindrical shaft arranged along the rotation axis Ax**1** on the input side. A central axis of the shaft used as the input portion **103** is consistent with the rotation axis Ax**1**. The input portion **103** is held by the input-side housing **111** to be rotatable with the rotation axis Ax**1** as a center. The input portion **103** is mounted on the cam **41** of the wave generator **4** and rotates with the rotation axis Ax**1** as the center, together with the cam **41**. That is, the input portion **103** transmits the power (rotation force) generated by the driving source **101** to the cam **41** as input. As described above, in some embodiments, the rotation axis Ax**1** on the input side and the rotation axis Ax**2** on the output side are on the same line, so that the input portion **103** and the output portion **102** are located on the same axis.
(93) The input-side housing **111** holds the input portion **103** via the input-side bearings **118**, **119**, so that the input portion **103** is rotatable. A pair of input-side bearings **118**, **119** are arranged at an interval there-between along the rotation axis Ax**1**. In some embodiments, the shaft used as the input portion **103** penetrates the input-side housing **111**, and a front end portion of the input portion **103** protrudes from an end surface (right end surface of FIG. **4**) of the input-side housing **111** at the input side of the rotation axis Ax**1**. A gap between the end surface of the input-side housing **111** at the input side of the rotation axis Ax**1** and the input portion **103** is blocked by the input-side oil seal **120**.
(94) The output-side housing **112** holds the output portion **102** via the output-side bearings **121**, **122**, so that the output portion **102** is rotatable. A pair of output-side bearings **121**, **122** are arranged

at an interval there-between along the rotation axis Ax**2**. In some embodiments, the shaft used as the output portion **102** penetrates the output-side housing **112**, and a front end portion of the output portion **102** protrudes from an end surface (left end surface of FIG. **4**) of the output-side housing **112** at the output side of the rotation axis Ax**1**. A gap between the end surface of the output-side housing **112** at the output side of the rotation axis Ax**1** and the output portion **102** is blocked by the output-side oil seal **123**.
(95) Here, as shown in FIG. **4**, the input-side housing **111** and the output-side housing **112** are combined with each other in a state where the rigid internal gear **2** of the harmonic gear device **1** is clamped from both sides in the direction parallel to the rotation axis Ax**1**, i.e., the tooth direction D**1**. Specifically, the input-side housing **111** is in contact with the rigid internal gear **2** from the input side of the rotation axis Ax**1**, and the output-side housing **112** is in contact with the rigid internal gear **2** from the output side of the rotation axis Ax**1**. In this way, in a state where the rigid internal gear **2** is clamped between the input-side housing **111** and the output-side housing **112**, the input-side housing **111** is firmly fixed to the output-side housing **112** by screws (bolts) passing through multiple fixing holes **22**. Therefore, the input-side housing **111**, the output-side housing **112** and the rigid internal gear **2** are combined with each other to form one piece. In other words, the rigid internal gear **2** constitutes outline of the actuator **100**, together with the input-side housing **111** and the output-side housing **112**.
(96) The spline bushing **113** is a cylindrical component configured to connect the shaft used as the input portion **103** to the cam **41**. The spline bushing **113** is inserted into the cam hole **43** formed in the cam **41**, and the shaft used as the input portion **103** is inserted into the spline bushing **113** by penetrating the spline bushing **113**. Here, movement of the spline bushing **113** relative to both the cam **41** and the input portion **103** is limited in a rotation direction with the rotation axis Ax**1** as the center, and the spline bushing **113** is at least movable relative to the input portion **103** in the direction parallel to the rotation axis Ax**1**. Therefore, it is possible to achieve a spline connection structure used as a connection structure of the input portion **103** and the cam **41**. Therefore, the cam **41** may move along the rotation axis Ax**1** relative to the input portion **103** and rotate with the rotation axis Ax**1** as the center, together with the input portion **103**.
(97) The spacer **114** is a component configured to fill a gap between the spline bushing **113** and the cam **41**. The first stopper **115** is a component configured to prevent the spline bushing **113** from falling away from the cam **41**. The first stopper **115** is for example composed of an E-ring and mounted at a position of the spline bushing **113** at the input side of the rotation axis Ax**1** with respect to the cam **41**. The second stopper **116** is a component configured to prevent the input portion **103** from falling away from the spline bushing **113**. The second stopper **116** is for example composed of an E-ring and mounted on the input portion **103** to be in contact with the spline bushing **113** from the output side of the rotation axis Ax**1**.
(98) The mounting plate **117** is a component configured to mount the shaft used as the output portion **102** to the bottom portion **322** of the flexible external gear **3**. Specifically, the mounting plate **117** is firmly fixed to a flange portion of the output portion **102** by screws (bolts) passing through multiple mounting holes **33**, in a state where portions around the through-hole **34** of the bottom portion **322** are clamped between the mounting plate **117** and the flange portion. Therefore, the shaft used as the output portion **102** is fixed to the bottom portion **322** of the flexible external gear **3**.
(4) Structures of Internal Tooth and External Tooth
(99) Next, structures of the internal tooth **21** and the external tooth **31** of the harmonic gear device **1** of the embodiment is described in more detail with reference to FIG. **5**A to FIG. **8**B.
(100) FIG. **5**A is a cross-sectional view observed in view of the internal tooth **21** and the external tooth **31** of FIG. **1**B. FIG. **5**B is a cross-sectional view of a line A**1**-A**1** of FIG. **5**A. FIG. **6** is a conceptual explanation view showing trimming amounts Q**1**, Q**2** of the internal tooth **21** and the external tooth **31**, and showing a state where meshing of the internal tooth **21** and the external tooth

31 is released from the state shown in FIG. **5**A. FIG. **7**A is a cross-sectional view of a line n**1**-n**1** of FIG. **5**A. FIG. **7**B is a cross-sectional view of a line B**2**-B**2** of FIG. **5**A. FIG. **7**C is a cross-sectional view of a line B**3**-B**3** of FIG. **5**A. FIG. **8**A is a cross-sectional view showing a tapered surface **302**, inclined with respect to the rotation axis Ax**1**, in the inner peripheral surface **301** of the flexible external gear **3**. FIG. **8**B is an enlarged view of a region Z**1** of FIG. **8**A. As described above, drawings referred by some embodiments of the disclosure are schematic views, and respective ratios of sizes and thicknesses of structural elements in the drawings do not necessarily reflect actual size ratios. Therefore, in FIG. **5**A to FIG. **7**C for example, trimming amounts Q**1**, Q**2** of tooth-direction trimming are represented by large values respectively, which are not aimed to limit actual shapes of the internal tooth **21** and the external tooth **31**. Furthermore, cross-sectional lines (oblique lines) of cross-sections are omitted in FIG. **5**A to FIG. **7**C.
(101) (4.1) Surface Hardness
(102) Firstly, surface hardness of the internal tooth **21** and the external tooth **31** in the embodiment are described.
(103) In some embodiments, as described above, surface hardness of the internal tooth **21** is lower than that of the external tooth **31**. That is, hardness of a surface of the external tooth **31** is higher (harder) than that of a surface of the internal tooth **21**. “hardness” stated in some embodiments of the disclosure refers to a hard degree of an object, and hardness of a metal is represented for example by size of an indentation formed by pressing a steel ball at a certain pressure. Specifically, as an example of hardness of the metal, there are Rockwell hardness (HRC), Brinell hardness (HB), Vickers hardness (HV), or Shore hardness (HS), or the like. In some embodiments, hardness is represented with Vickers hardness (HV) as long as there is no specific limitation. As a means of increasing hardness (hardening) of a metal part, there are for example alloying, or heat treatment, or the like.
(104) In some embodiments, the surface of the external tooth **31** of the flexible external gear **3** is made of a material having high hardness and high flexibility (toughness), and the internal tooth **21** of the rigid internal gear **2** is made of a material having a lower hardness than that of the external tooth **31**. In some embodiments, as an example, the external tooth **31** use a material obtained by thermal treatment (quenching and tempering) of a nickel-chrome molybdenum steel specified by the Japanese Industrial Standards (JIS) as “SNCM 439”. The internal tooth **21** uses a spherical graphite cast iron specified by JIS as “FCD800-2”.
(105) Furthermore, surface hardness of the internal tooth **21** which is relatively low hardness compared with the external tooth **31** is preferably less than HV350. In some embodiments, as an example, surface hardness of the internal tooth **21** is selected in a range of greater than HV250 and less than HV350. A lower limit of surface hardness of the internal tooth **21** is not limited to HV250, for example, may be HV150, HV160, HV170, HV180, HV190, HV200, HV210, HV220, HV230, or HV240, or the like. Similarly, an upper limit of surface hardness of the internal tooth **21** is not limited to HV350, for example, may be HV360, HV370, HV380, HV390, HV400, HV410, HV420, HV430, HV440, or HV450, or the like.
(106) In comparison, surface hardness of the external tooth **31** which is relatively high hardness compared with the internal tooth **21** is preferably greater than HV380. In some embodiments, as an example, surface hardness of the external tooth **31** is selected in a range of greater than HV380 and less than HV450. A lower limit of surface hardness of the external tooth **31** is not limited to HV380, for example, may be HV280, HV290, HV300, HV310, HV320, HV330, HV340, HV350, HV360, or HV370, or the like. Similarly, an upper limit of surface hardness of the internal tooth **21** is not limited to HV450, for example, may be HV460, HV470, HV480, HV490, HV500, HV510, HV520, HV530, HV540, or HV550, or the like.
(107) Furthermore, in some embodiments, a difference between surface hardness of the internal tooth **21** and surface hardness of the external tooth **31** is greater than HV50. That is, surface hardness of the external tooth **31** is set to be higher than that of the internal tooth **21** by greater than

HV50. In summary, for example when surface hardness of the internal tooth **21** is HV350, surface hardness of the external tooth **31** is greater than HV400. Furthermore, when surface hardness of the external tooth **31** is HV380, surface hardness of the internal tooth **21** is less than HV330. The difference between surface hardness of the internal tooth **21** and surface hardness of the external tooth **31** is not limited to greater than HV50, for example, may be greater than HV20, greater than HV30, or greater than HV40. Furthermore, it is more preferable when the difference between surface hardness of the internal tooth **21** and surface hardness of the external tooth **31** is larger, for example, the difference is more preferably greater than HV60, greater than HV70, greater than HV80, greater than HV90, or greater than HV100. In case that the difference between surface hardness of the internal tooth **21** and surface hardness of the external tooth **31** is greater than HV100, when surface hardness of the internal tooth **21** is HV350, surface hardness of the external tooth **31** is greater than HV450.

(108) As described above, in some embodiments, surface hardness of the internal tooth **21** is set to lower than that of the external tooth **31**. Therefore, when the harmonic gear device **1** operates, the internal tooth **21** with relatively low surface hardness is actively worn compared with the external tooth **31** when the internal tooth **21** is in contact with the external tooth **31**. When two components with different surface hardness (internal tooth **21** and external tooth **31**) are in contact with each other, wear of the relatively soft internal tooth **21** proceeds, which may suppress wear of the relatively hard external tooth **31**. That is, during an initial stage of usage of the harmonic gear device **1**, a tooth surface of the internal tooth **21** is appropriately worn, so that a real contact area between the internal tooth **21** and the external tooth **31** is enlarged, and a surface pressure is reduced, so that it is difficult to generate wear of the external tooth **31**. Furthermore, when surface hardness of the internal tooth **21** is less than HV350 as in some embodiments, even when the internal tooth **21** is in contact with the external tooth **31**, a foreign matter X**1** is generated due to absence or wear of the internal tooth **21** or the like, while the foreign matter X**1** is relatively soft. In summary, the foreign matter X**1** which is easily generated due to wear during the initial stage of usage of the harmonic gear device **1** is a soft foreign matter X**1** generated from the relatively soft internal tooth **21**, and thus, for example even when the foreign matter X**1** enters the bearing **42**, damage to the bearing **42** may be suppressed. As a result, for example, generation amount or the like of a hard foreign matter X**1** which damages the bearing **42** greatly may be suppressed. Especially when the difference between surface hardness of the internal tooth **21** and surface hardness of the external tooth **31** is a large value such as greater than HV **50**, the above effect is significant.

(109) Furthermore, an effect of suppressing sintering of tooth surfaces of the internal tooth **21** and the external tooth **31** during an initial stage of wear of the internal tooth **21** may be expected by using the spherical graphite cast iron as material of the internal tooth **21**. Therefore, a lubrication effect of the meshing position of the internal tooth **21** and the external tooth **31** may be obtained, and power transmission efficiency of the harmonic gear device **1** may be improved.

(110) Surface hardness of the internal tooth **21** and of the external tooth **31** are not necessarily specified by Vickers hardness (HV), and surface hardness of the internal tooth **21** and of the external tooth **31** may also be specified by other hardness, such as Rockwell hardness (HRC), Brinell hardness (HB), or Shore hardness (HS).

(111) (4.2) Tooth-Direction Trimming

(112) Next, tooth-direction trimming of the internal tooth **21** and the external tooth **31** of the embodiment are described.

(113) As a premise, as shown in FIG. **5**A, the internal tooth **21** has a tooth bottom **212** and a tooth crest **213**. The internal tooth **21** is arranged on the inner peripheral surface of the rigid internal gear **2**, therefore, the tooth bottom **212** of the internal tooth **21** is equivalent to the inner peripheral surface of the rigid internal gear **2**, and the tooth crest **213** protrudes from the inner peripheral surface of the rigid internal gear **2** toward an inner side (center of the rigid internal gear **2**).

(114) On the other hand, as shown in FIG. **5**A, the external tooth **31** has a tooth bottom **312** and a tooth crest **313**. The external tooth **31** is arranged on the outer peripheral surface of (the trunk portion **321** of) the flexible external gear **3**, therefore, the tooth bottom **312** of the external tooth **31** is equivalent to the outer peripheral surface of (the trunk portion **321** of) the flexible external gear **3**, and the tooth crest **313** protrudes from the outer peripheral surface of (the trunk portion **321** of) the flexible external gear **3** toward an outer side.
(115) At the meshing position of the internal tooth **21** and the external tooth **31**, the tooth crest **313** of the external tooth **31** is inserted between an adjacent pair of tooth crests **213** of the internal tooth **21**, so that the internal tooth **21** is mesh with the external tooth **31**. At this time, the tooth crest **313** of the external tooth **31** is opposite to the tooth bottom **212** of the internal tooth **21**, and the tooth crest **213** of the internal tooth **21** is opposite to the tooth bottom **312** of the external tooth **31**. Furthermore, a desirable situation is to ensure a slight gap between the tooth bottom **212** of the internal tooth **21** and the tooth crest **313** of the external tooth **31** and a slight gap between the tooth bottom **312** of the external tooth **31** and the tooth crest **213** of the internal tooth **21**. In this state, tooth surfaces of the internal tooth **21** and the external tooth **31** opposite to each other in a tooth-thickness direction D**2** (referring to FIG. **5**B) are in contact with each other to perform power transmission between the rigid internal gear **2** and the flexible external gear **3**.
(116) Furthermore, chamfered portions **211** are provided at both end portions in the tooth direction D**1** of the internal tooth **21**. The chamfered portion **211** is a C-shaped surface reducing amount of protrusion of the internal tooth **21** toward both sides in the tooth direction D**1**, which is a component not taking effect on meshing of the internal tooth **21** and the external tooth **31** substantially. That is, the chamfered portion **211** of the internal tooth **21** is not connected to the external tooth **31** even at the meshing position of the internal tooth **21** and the external tooth **31**. Similarly, chamfered portions **311** are provided at both end portions in the tooth direction D**1** of the external tooth **31**. The chamfered portion **311** is a C-shaped surface reducing amount of protrusion of the internal tooth **21** toward both sides in the tooth direction D**1**, which is a component not taking effect on meshing of the internal tooth **21** and the external tooth **31** substantially. That is, the chamfered portion **311** of the external tooth **31** is not connected to the internal tooth **21** even at the meshing position of the internal tooth **21** and the external tooth **31**.
(117) Here, in some embodiments, as shown in FIG. **5**A, FIG. **5**B and FIG. **6**, the internal tooth **21** of the rigid internal gear **2** has a tooth-direction trimming portion **210**. That is, as to the harmonic gear device **1**, tooth-direction trimming is performed on at least the internal tooth **21**. The tooth-direction trimming portion **210** of the internal tooth **21** is arranged at an end portion in at least one side along the tooth direction D**1**. In other words, the internal tooth **21** has a tooth-direction trimming portion **210** at an end portion in at least one side along the tooth direction D**1** of the internal tooth **21**. In some embodiments, the tooth-direction trimming portion **210** is arranged at both end portions in the tooth direction D**1** of the internal tooth **21**.
(118) Furthermore, in some embodiments, the external tooth **31** of the flexible external gear **3** also has a tooth-direction trimming portion **310**. That is, as to the harmonic gear device **1**, tooth-direction trimming is performed on not only the internal tooth **21** but also the external tooth **31**. The tooth-direction trimming portion **310** of the external tooth **31** is arranged at an end portion in at least one side along the tooth direction D**1**. In other words, the external tooth **31** has a tooth-direction trimming portion **310** at an end portion in at least one side along the tooth direction D**1** of the external tooth **31**. In some embodiments, the tooth-direction trimming portion **310** is arranged at both end portions in the tooth direction D**1** of the external tooth **31**.
(119) In this way, in the harmonic gear device **1** of some embodiments, at least one of the internal tooth **21** or the external tooth **31** is provided with tooth-direction trimming portions **210**, **310**. Therefore it is difficult to generate stress concentration induced by excessive tooth contact between the internal tooth **21** and the external tooth **31**, due to the tooth-direction trimming portions **210**, **310**. As a result, tooth contact of the internal tooth **21** and the external tooth **31** may be improved.

Therefore it is difficult to generate the foreign matter X**1** due to absence, or wear, or the like induced by contact between the internal tooth **21** and the external tooth **31**, which may provide the harmonic gear device **1** with high reliability.
(120) Here the tooth-direction trimming portion **210** is arranged on at least the internal tooth **21**. The tooth-direction trimming portion **210** is arranged on the rigid internal gear **2** (the internal tooth **21**), so that tooth-direction trimming is unnecessary for the flexible external gear **3** (the external tooth **31**), or a trimming amount of the flexible external gear **3** may be reduced, and reduction of strength of the flexible external gear **3** induced by performing tooth-direction trimming on the flexible external gear **3** with flexibility is easily suppressed. That is, the flexible external gear **3** is formed by a thin-wall metal elastomer (metal plate) as described above, therefore, the flexible external gear **3** has flexibility due to its small (thin) thickness. Therefore, when excessive tooth-direction trimming is performed on the external teeth **31** of the flexible external gear **3**, the flexible external gear **3** which is thin originally will become thinner, possibly resulting in reduction of strength of the flexible external gear **3**. Especially when excessive tooth-direction trimming is performed on the tooth bottom **312** of the external teeth **31**, it is difficult to ensure thickness required by the trunk portion **321** to maintain strength. In comparison, in some embodiments, the tooth-direction trimming portion **210** is arranged on the internal tooth **21**, so that a trimming amount may be reduced for tooth-direction trimming of the flexible external gear **3**. As a result, it is easy to maintain strength of the flexible external gear **3**.
(121) Furthermore, in some embodiments, not only the internal tooth **21** but also the external tooth **31**, that is, both the internal tooth **21** and the external tooth **31** are provided with tooth-direction trimming portions **210**, **310**. Therefore, excessive tooth-direction trimming is unnecessary for the internal tooth **21**, and desired performance may be achieved by tooth-direction trimming with a moderate trimming amount. In some embodiments of the disclosure, in case of distinguishing the tooth-direction trimming portion **210** of the internal tooth **21** from the tooth-direction trimming portion **310** of the external tooth **31**, the tooth-direction trimming portion **210** of the internal tooth **21** is also referred to as a “first trimming portion **210**”, and the tooth-direction trimming portion **310** of the external tooth **31** is referred to as a “second trimming portion **310**”. That is, in some embodiments, the internal tooth **21** has a first trimming portion **210** used as the tooth-direction trimming portion **210**. The external tooth **31** has a second trimming portion **310** which is a different tooth-direction trimming portion **310** from the first trimming portion **210**.
(122) As shown in FIG. **6**, in case of performing tooth-direction trimming on both the internal tooth **21** and the external tooth **31**, a sum of the trimming amount Q**1** of the internal tooth **21** and the trimming amount Q**2** of the external tooth **31** may be considered as a trimming amount of tooth-direction trimming of the internal tooth **21** and the external tooth **31**. Therefore, when the trimming amount of tooth-direction trimming of the internal tooth **21** is the same as that of tooth-direction trimming of the external tooth **31**, the trimming amount Q**1** of the internal tooth **21** may be suppressed to reduce an amount of the trimming amount Q**2** of the external tooth **31**, compared with a case that tooth-direction trimming is performed only on the internal tooth **21**. In FIG. **6**, a regular tooth-direction shape is represented by an imaginary line (double dotted lines), and a maximum shift amount implemented by tooth-direction trimming with respect to the regular tooth-direction shape is expressed as trimming amounts Q**1**, Q**2**. That is, as shown in FIG. **6**, the trimming amount Q**1** of the internal tooth **21** is a shift amount with respect to a central portion of the tooth crest **213** in the tooth direction D**1** (a tooth-height direction). Similarly, the trimming amount Q**2** of the external tooth **31** is a shift amount with respect to a central portion of the tooth crest **313** in the tooth direction D**1** (a tooth-height direction). In FIG. **6**, the trimming amounts Q**1**, Q**2** of the tooth crests **213**, **313** are exemplified, however, the trimming amounts are not limited to the tooth crests **213**, **313**, and also applicable to trimming amounts of the tooth bottoms **212**, **312** or at an end surface in the tooth-thickness direction D**2**.
(123) However, as described above, representative processes of tooth-direction trimming include

drum-shaped trimming and edge trimming process. In some embodiments, the tooth-direction trimming portion (the first trimming portion **210**) of the internal tooth **21** is formed by drum-shaped trimming. That is, the first trimming portion **210** is formed by processing the internal tooth **21** to allow the internal tooth **21** to have rounded corners facing toward a central portion in the tooth direction D**1** in a manner that the central portion in the tooth direction D**1** becomes a protrusion. Similarly, the tooth-direction trimming portion (the second trimming portion **310**) of the external tooth **31** is also formed by drum-shaped trimming. That is, the second trimming portion **310** is formed by processing the external tooth **31** to allow the external tooth **31** to have rounded corners facing toward a central portion in the tooth direction D**1** in a manner that the central portion in the tooth direction D**1** becomes a protrusion. In this way, in some embodiments, the first trimming portion **210** and the second trimming portion **310** include the same type of trimming. That is, both the first trimming portion **210** and the second trimming portion **310** are based on the same type of tooth-direction trimming (here drum-shaped trimming).

(124) Furthermore, the tooth-direction trimming portion (the first trimming portion **210**) of the internal tooth **21** allows at least one of the tooth bottom **212**, the tooth crest **213**, or an end surface in the tooth-thickness direction D**2** of the internal tooth **21** to include an inclined surface inclined with respect to the tooth direction D**1**. That is, the tooth-direction trimming portion **210** is formed by processing at least one of the tooth bottom **212**, the tooth crest **213**, or an end surface in the tooth-thickness direction D**2** of the internal tooth **21** for tooth-direction trimming (here drum-shaped trimming). When tooth-direction trimming is performed on the tooth bottom **212**, an inclined surface which is inclined with respect to the tooth direction D**1** in a manner that the tooth bottom **212** decreases with closing to both end portions in the tooth direction D**1**, is formed at the tooth bottom **212**. When tooth-direction trimming is performed on the tooth crest **213**, an inclined surface which is inclined with respect to the tooth direction D**1** in a manner that the tooth crest **213** decreases with closing to both end portions in the tooth direction D**1**, is formed at the tooth crest **213**. When tooth-direction trimming is performed on an end surface in the tooth-thickness direction D**2**, an inclined surface which is inclined with respect to the tooth direction D**1** in a manner that tooth thickness reduces with closing to both end portions in the tooth direction D**1**, is formed at the end surface in the tooth-thickness direction D**2**. In some embodiments, as shown in FIG. **5**A and FIG. **5**B, tooth-direction trimming is performed on the tooth bottom **212**, the tooth crest **213**, and both end surfaces in the tooth-thickness direction D**2** of the internal tooth **21**.

(125) In particular, the tooth-direction trimming portion **210** includes an inclined surface at least at the tooth bottom **212**, thereby easily avoiding interference between the tooth crest **313** of the external tooth **31** and the tooth bottom **212** of the internal tooth **21**. That is, when tooth-direction trimming is performed on the tooth bottom **212** of the internal tooth **21**, it may ensure a gap for an "avoidance portion" of the tooth crest **313** of the external tooth **31**, and it is difficult to generate excessive stress concentration induced by interference between the tooth crest **313** of the external tooth **31** and the tooth bottom **212** of the internal tooth **21**.

(126) The above principle is also applicable to the external tooth **31**. The tooth-direction trimming portion (the second trimming portion **310**) allows at least one of the tooth bottom **312**, the tooth crest **313**, or an end surface in the tooth-thickness direction D**2** of the external tooth **31** to include an inclined surface inclined with respect to the tooth direction D**1**. That is, the tooth-direction trimming portion **310** is formed by processing at least one of the tooth bottom **312**, the tooth crest **313**, or an end surface in the tooth-thickness direction D**2** of the external tooth **31** for tooth-direction trimming (here drum-shaped trimming). In some embodiments, as shown in FIG. **5**A and FIG. **5**B, tooth-direction trimming is performed on the tooth bottom **312**, the tooth crest **313**, and both end surfaces in the tooth-thickness direction D**2** of the external tooth **31**.

(127) As shown in FIG. **5**B, by performing tooth-direction trimming on both end surfaces in the tooth-thickness direction D**2**, tooth thickness of each of the internal tooth **21** and the external tooth **31** is maximal at a central portion in the tooth direction D**1**, and gradually reduces toward both end

portions in the tooth direction D**1**. Therefore, in terms of the meshing position of the internal tooth **21** and the external tooth **31**, a gap between the internal tooth **21** and the external tooth **31** is smallest at the central portion in the tooth direction D**1** on which tooth-direction trimming (here drum-shaped trimming) is not performed. A portion in the tooth direction D**1** of each of the internal tooth **21** and the external tooth **31** on which tooth-direction trimming is not performed, i.e. a portion still having a regular tooth-direction shape, is also referred to as a “non-trimming portion”. Therefore, in terms of the meshing position of the internal tooth **21** and the external tooth **31**, “non-trimming portions” of central portions of the internal tooth **21** and the external tooth **31** in the tooth direction D**1** are substantially in contact with each other at first.

(128) In summary, in examples of FIG. **7**A to FIG. **7**C, in FIG. **7**A showing a cross-section of a central portion in the tooth direction D**1**, a gap G**1** between the internal tooth **21** and the external tooth **31** is minimal. That is, the gap G**1** between the internal tooth **21** and the external tooth **31** becomes larger toward an end portion in the tooth direction D**1** (the output side of the rotation axis Ax**1**), according to a sequence of FIG. **7**B and FIG. **7**C. In this way, by tooth-direction trimming, the farther the tooth surface away from the central portion in the tooth direction D**1**, the greater the tooth surface shifting in a negative direction gradually, therefore, the gap G**1** between the internal tooth **21** and the external tooth **31** becomes larger.

(129) In some embodiments, such non-trimming portion (the central portion in the tooth direction D**1**) is arranged at a repetitive position with the rolling body **423** of the bearing **42** in the tooth direction D**1**. Strictly, the non-trimming portion (the central portion in the tooth direction D**1**) is located on a line (imaginary line) orthogonal to the rotation axis Ax**1** and passing through center of the rolling body **423**. Therefore, stress transmitted from multiple rolling bodies **423** to the flexible external gear **3** via the outer ring **421** of the bearing **42** mainly acts on the non-trimming portion, so that the internal tooth **21** and the external tooth **31** are easily in contact with each other at the non-trimming portion at first.

(130) Furthermore, in some embodiments, there is a difference between trimming amounts Q**1**, Q**2** of the first trimming portion **210** of the internal tooth **21** and the second trimming portion **310** of the external tooth **31**. The trimming amount Q**2** of the second trimming portion **310** is less than the trimming amount Q**1** of the first trimming portion **210** (Q**1**>Q**2**). The flexible external gear **3** has a function of elastic deformation (deflection) different from a power transmission function, and stress generated by deflection is inherent, so that the external teeth **31** of the flexible external gear **3** are required to tolerate bending stress. Furthermore, when tooth-direction trimming is performed on the external tooth **31**, a tooth root of the external tooth **31** become thin substantially, therefore tolerance of the bending stress decreases, and durability of the flexible external gear **3** decreases. On the other hand, elastic deformation is unnecessary for the rigid internal gear **2**, so that the internal teeth **21** of the rigid internal gear **2** are not required to tolerate bending stress. Therefore, durability of the rigid internal gear **2** is hardly affected, even when tooth-direction trimming is performed on the internal tooth **21** so that a tooth root of the internal tooth **21** becomes thin. Therefore, by making the trimming amount Q**2** of the external tooth **31** less than the trimming amount Q**1** of the internal tooth **21**, it may ensure sufficient trimming amounts for tooth-direction trimming performed on the internal tooth **21** and the external tooth **31** and suppress decrease in durability of the harmonic gear device **1**.

(131) Furthermore, the tooth-direction trimming portion **210** is arranged at both end portions in the tooth direction D**1** of the internal tooth **21**. In other words, the tooth-direction trimming portion **210** is arranged at end portions of the internal tooth **21** at both input and output sides of the rotation axis Ax**1**. That is, the tooth-direction trimming portion **210** is arranged at an end portion in at least the opened surface **35** side along the tooth direction D**1** of the internal tooth **21**. Here, in a state where the flexible external gear **3** is elastically deformed, an end portion at the opened surface **35** side of the flexible external gear **3** in the direction of the rotation axis Ax**1** is deformed greater than deformation of an end portion at the bottom portion **322** side, to form a shape closer to an elliptical

shape. Due to such difference in amount of deformation in the direction of the rotation axis Ax**1**, a tapered surface **302** (referring to FIG. **8**A) inclined with respect to the rotation axis Ax**1** is generated on the inner peripheral surface **301** of the trunk portion **321** of the flexible external gear **3**, in a state where the flexible external gear **3** is elastically deformed. Furthermore, by providing the tooth-direction trimming portion **210** at the opened surface **35** side along the tooth direction D**1**, tooth-direction trimming may be performed on the internal tooth **21** to avoid (evade) the external tooth **31** which is inclined due to the tapered surface **302**. Therefore it may be difficult to generate stress concentration through the tooth-direction trimming portion **210**, even at the end portion at the opened surface **35** side along the tooth direction D**1** where stress concentration is generated very easily due to deformation of the external tooth **31**.
(132) Furthermore, the tooth-direction trimming portion (the first trimming portion **210**) of the internal tooth **21** is formed for example by a rotary scraping cutter. That is, the tooth-direction trimming portion **210** may be formed by performing tooth-direction trimming using the rotary scraping cutter on the internal teeth **21** formed on the inner peripheral surface of the rigid internal gear **2**. The rotary scraping cutter has multiple small-gear-shaped cutting edges which form gear portions (the internal teeth **21**) in a workpiece (the rigid internal gear **2**). The rotary scraping cutter is driven to rotate with a cutter axle core different from a workpiece axle core as a center, then the rotary scraping cutter rotates synchronously with the workpiece while cuts the workpiece by relatively moving in the tooth direction D**1**. Therefore gear portions (the internal teeth **21**) on which tooth-direction trimming is performed, are formed in the workpiece (the rigid internal gear **2**).
(133) Furthermore, in some embodiments, not only the above tooth-direction trimming is performed on the internal tooth **21** and the external tooth **31**, but also tooth-shape trimming is performed on the internal tooth **21** and the external tooth **31** as shown in FIG. **7**A to FIG. **7**C. With respect to tooth-shape trimming, in a state where the flexible external gear **3** in an elastically deformed state is arranged on the inner side of the rigid internal gear **2** and does not perform power transmission, a trimming amount of tooth-shape trimming is determined in a manner that the following conditions are satisfied. That is, the conditions are that a gap is not generated near a pitch point of the meshing position of the internal tooth **21** and the external tooth **31**, and a gap which is greater than for example 0.1 modulus is generated between the tooth crest **313** of the external tooth **31** and the tooth bottom **212** of the internal tooth **21**. In order to meet such conditions, tooth-shape trimming is performed on each of the internal tooth **21** and the external tooth **31**, to have a shape with rounded corners facing toward a central portion in a tooth-height direction in a manner that the central portion in the tooth-height direction becomes a protrusion. Therefore, even when the internal teeth **21** of the rigid internal gear **2** are worn, contact with the tooth bottom is difficult to occur.
(134) (4.3) Tooth Width
(135) Next, tooth widths (sizes in the tooth direction D**1**) of the internal tooth **21** and the external tooth **31** of the embodiment are described.
(136) In some embodiments, as shown in FIG. **5**A, the external tooth **31** of the flexible external gear **3** protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21** of the rigid internal gear **2**. That is, as described above, in the external teeth **31** of the flexible external gear **3** and the internal teeth **21** of the rigid internal gear **2**, central positions thereof in the tooth direction D**1** is aligned with the same position in the direction of the rotation axis Ax**1**. Furthermore, the size (tooth width) of the external tooth **31** in the tooth direction D**1** is greater than the size (tooth width) of the internal tooth **21** in the tooth direction D**1**. Therefore, in the tooth direction D**1**, the internal tooth **21** is received in a range of the external tooth **31** in the tooth direction and the external tooth **31** protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21**.
(137) Especially in some embodiments, as shown in FIG. **5**A, the external tooth **31** protrudes toward both sides along the tooth direction D**1** (input and output sides of the rotation axis Ax**1**)

relative to the internal tooth **21**. The external tooth **31** protrudes from an end edge of the internal tooth **21** toward one side along the tooth direction D**1** (the input side of the rotation axis Ax**1**) by a protrusion amount L**1**. The external tooth **31** protrudes from an end edge of the internal tooth **21** toward the other side along the tooth direction D**1** (the output side of the rotation axis Ax**1**) by a protrusion amount L**2**. In some embodiments, as an example, protrusion amounts L**1**, L**2** by which the external tooth **31** protrudes from the internal tooth **21** are substantially the same at both sides along the tooth direction D**1**.

(138) Here protrusion amounts L**1**, L**2** are represented by comparison between a portion of the internal tooth **21** other than the chamfered portion **211** and a portion of the external tooth **31** other than the chamfered portion **311**. That is, at one side along the tooth direction D**1** (the input side of the rotation axis Ax**1**), a distance from a starting point of the chamfered portion **211** of the internal tooth **21** to a starting point of the chamfered portion **311** of the external tooth **31** is the protrusion amount L**1**. Similarly, at the other side along the tooth direction D**1** (the output side of the rotation axis Ax**1**), a distance from a starting point of the chamfered portion **211** of the internal tooth **21** to a starting point of the chamfered portion **311** of the external tooth **31** is the protrusion amount L**2**.

(139) Furthermore, as described in a "(4.1) Surface Hardness" column, surface hardness of the internal tooth **21** is lower than that of the external tooth **31**. Furthermore, in some embodiments, the external tooth **31** with relatively high surface hardness protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21**, therefore, in at least one side along the tooth direction D**1**, it is difficult to generate a height difference induced by wear at a tooth surface of the internal tooth **21**. That is, in at least one side along the tooth direction D**1**, the internal tooth **21** with relatively low surface hardness is evenly worn due to tooth contact between it and the external tooth **31**, so that it is difficult to generate a local recess (height difference) at the tooth surface of the internal tooth **21**. Therefore, even though there is a situation where a tooth contact position deviates in the tooth direction D**1** due to certain jitter, occurrence of abnormality of the harmonic gear device **1** due to an excessive load acting at the meshing position of the internal tooth **21** and the external tooth **31** is easily suppressed. Therefore it is difficult to generate the foreign matter X**1** due to absence, or wear, or the like induced by contact between the internal tooth **21** and the external tooth **31**, and the harmonic gear device **1** with high reliability may be provided.

(140) Furthermore, the external tooth **31** protrudes toward both sides along the tooth direction D**1** relative to the internal tooth **21**. That is, the external tooth **31** protrudes toward at least the opened surface **35** side along the tooth direction D**1** relative to the internal tooth **21**. Here, in a state where the flexible external gear **3** is elastically deformed, an end portion at the opened surface **35** side of the flexible external gear **3** in the direction of the rotation axis Ax**1** is deformed greater than deformation of an end portion at the bottom portion **322** side, to form a shape closer to an elliptical shape. Due to such difference between deformation amounts in the direction of the rotation axis Ax**1**, a tapered surface **302** (referring to FIG. **8**A) inclined with respect to the rotation axis Ax**1** is generated on the inner peripheral surface **301** of the trunk portion **321** of the flexible external gear **3**, in a state where the flexible external gear **3** is elastically deformed. Furthermore, the external tooth **31** protrudes from the internal tooth **21** at the opened surface **35** side along the tooth direction D**1**, thereby avoiding a corner portion at a front end of the external tooth **31** inclined due to such tapered surface **302** from contacting the internal tooth **21**. Therefore, it is difficult to generate the local recess (height difference) at the tooth surface of the internal tooth **21**, even at the end portion at the opened surface **35** side along the tooth direction D**1** where stress concentration is generated very easily due to deformation of the external tooth **31**.

(141) (4.4) Tapered Surface

(142) Next, the tapered surface **302** generated on the inner peripheral surface **301** of the flexible external gear **3**, which is equivalent to an inner side of the external tooth **31**, is described.

(143) As described above, in a state where the flexible external gear **3** is elastically deformed, an end portion at the opened surface **35** side of the flexible external gear **3** in the direction of the

rotation axis Ax**1** is deformed greater than deformation of an end portion at the bottom portion **322** side, to form a shape closer to an elliptical shape. Therefore, as shown in FIG. **8**A and FIG. **8**B, in a state where the flexible external gear **3** is elastically deformed, the inner peripheral surface **301** of the trunk portion **321** of the flexible external gear **3** includes a tapered surface **302** inclined with respect to the rotation axis Ax**1**.
(144) The tapered surface **302** is inclined by an inclination angle θ**1** with respect to the rotation axis Ax**1**. Since such tapered surface **302** is generated, a gap between an outer peripheral surface **424** (referring to FIG. **8**B) of the outer ring **421** of the bearing **42** of the wave generator **4** embedded in the inner side of the trunk portion **321** and the inner peripheral surface **301** (the tapered surface **302**) gradually increases toward the opened surface **35** side. In summary, the inner peripheral surface **301** of the flexible external gear **3** has a tapered surface **302** at a position opposite to the outer peripheral surface **424** (of the outer ring **421** of the bearing **42**) of the wave generator **4**, and the tapered surface **302** increases a gap between it and the outer peripheral surface **424** of the wave generator **4** in a direction (the open surface **35** side) along the rotation axis Ax**1**.
(145) Here an inclination angle θ**1** of the tapered surface **302** with respect to the rotation axis Ax**1** is less than 5 degrees. Therefore, the gap generated between the tapered surface **302** and the outer peripheral surface **424** of the wave generator **4** is a small gap. In some embodiments, the small gap generated between the tapered surface **302** and the outer peripheral surface **424** of the wave generator **4** is used to maintain the lubricant Lb**1**. Specifically, a liquid or gel-like lubricant Lb**1** may be maintained in the small gap between the tapered surface **302** and the outer peripheral surface **424** of the wave generator **4**.
(146) That is, in the harmonic gear device **1** of some embodiments, the liquid or gel-like lubricant Lb**1** is injected for example in a meshing portion of the internal tooth **21** and the external tooth **31**, between the outer ring **421** and the inner ring **422** of the bearing **42**, or the like. As an example, the lubricant Lb**1** is a liquid lubrication oil (oil). Furthermore, as shown in FIG. **8**B, when the harmonic gear device **1** is used, the lubricant Lb**1** also enters between the outer ring **421** (the outer peripheral surface **424**) of the bearing **42** and the tapered surface **302** of the flexible external gear **3**. Therefore, the lubricant Lb**1** seals the gap between the outer peripheral surface **424** and the tapered surface **302**.
(147) Here, with respect to the bearing **42** of the wave generator **4**, a space at the input side of the rotation axis Ax**1** (right side of FIG. **8**A) may be connected to a space at the output side of the rotation axis Ax**1** (left side of FIG. **8**A) by the gap between the outer peripheral surface **424** and the tapered surface **302**. In this way, a gap forming a narrow path which connects an outer side (the input side of the rotation axis Ax**1**) to an inner side (the output side of the rotation axis Ax**1**) of the wave generator **4** is filled with the lubricant Lb**1**. Therefore, the space at the input side of the rotation axis Ax**1** and the space at the output side of the rotation axis Ax**1** are covered by the lubricant Lb**1**. Therefore, as shown in FIG. **8**B, the foreign matter X**1** entering from one side (input side) of the rotation axis Ax**1** to the inner side of the wave generator **4** is blocked by the lubricant Lb**1**.
(148) That is, in some embodiments, the lubricant Lb**1** is maintained in the gap between the tapered surface **302** and the outer peripheral surface **424** of the wave generator **4**. More specifically, as shown in FIG. **8**B, the lubricant Lb**1** is maintained in the small gap between the tapered surface **302** and the outer peripheral surface **424** of the wave generator **4** by a capillary phenomenon. Therefore, a state where the gap between the tapered surface **302** and the outer peripheral surface **424** of the wave generator **4** is filled with the lubricant Lb**1**, may be maintained. A magnitude of the holding force generated by the capillary phenomenon is also changed depending on “wettability” between the tapered surface **302**, the outer peripheral surface **424** and the lubricant Lb**1**. Therefore, it is preferable for at least the tapered surface **302** or the outer peripheral surface **424** not to have oleophobicity.
(149) The inclination angle θ**1** of the tapered surface **302** with respect to the rotation axis Ax**1** is

not limited to be less than 5 degrees, for example, it may be less than 10 degrees, less than 15 degrees, or less than 20 degrees. Furthermore, a case where the lubricant Lb**1** is maintained in the gap between the tapered surface **302** and the outer peripheral surface **424** of the wave generator **4**, is not a structure necessarily provided in the harmonic gear device **1**, or the lubricant Lb**1** may not be maintained in the gap.

(5) Function

(150) Next, functions of the harmonic gear device **1** of the embodiment are described in more detail.

(151) In a state where the harmonic gear device **1** elastically deforms the flexible external gear **3**, the internal teeth **21** are partially meshed with the external teeth **31**, so that meshing of the internal teeth **21** and the external teeth **31** is accompanied with “sliding” in both a tooth-shape direction (a direction orthogonal to the tooth direction D**1**) and the tooth direction D**1**. Furthermore, a flow rate of the lubricant Lb**1** is a relatively low speed in a rotation speed region of relative rotation of two gears (the rigid internal gear **2** and the flexible external gear **3**) when the harmonic gear device **1** is used normally. Therefore, at the meshing position of the internal tooth **21** and the external tooth **31**, flow of the lubricant Lb**1** with this relatively low speed may be generated in both the tooth-shape direction and the tooth direction D**1**, and the foreign matter X**1** generated between the internal tooth **21** and the external tooth **31** is difficult to flow to outside of the harmonic gear device **1**, through the lubricant Lb**1**. Therefore, as described above, the generated foreign matter X**1** is easily stopped in the harmonic gear device **1**, and reliability of the harmonic gear device **1** is reduced since such foreign matter X**1** enters the bearing **42** or the like to induce peeling with a type where points occur on surfaces.

(152) As a countermeasure against such reduction of reliability of the harmonic gear device **1**, a case where a side of the bearing **42** is blocked (sealed) by a sealing member to suppress entry of the foreign matter X**1** into the bearing **42**, may be considered. However, a sealing member used in the bearing **42** which has a relatively high reduction ratio and is elastically deformed, has a high friction loss and is difficult to prolong its service life, and reduction of power transmission efficiency becomes a problem. As other countermeasures, for example, performing thermal treatment with strong resistance against micropitting damage such as carburization nitriding treatment or the like on the outer ring **421** and the inner ring **422** of the bearing **42** to improve surface hardness of the outer ring **421** and the inner ring **422**, may be considered. However, there is a problem that surface hardness of the outer ring **421** and the inner ring **422** may not be improved too much, in order to avoid hindering elastic deformation of the bearing **42**.

(153) In comparison, in the harmonic gear device **1** of some embodiments, it is originally difficult to generate the foreign matter X**1**, thereby eliminating problems generated in the countermeasures for the bearing **42** as described above. That is, in the harmonic gear device **1**, suppression of generation of the foreign matter X**1** as in the embodiment is particularly useful itself, thereby having advantages that not only it is difficult to reduce reliability of the harmonic gear device **1**, but also it is easy to prolong the service life and improve the power transmission efficiency.

(154) As an example, in case that peeling due to entry of the foreign matter X**1** occurs on the inner peripheral surface (rolling surface) of the outer ring **421**, functions of the bearing **42** are hindered, and a fault may occur for operations of the harmonic gear device **1**. In the harmonic gear device **1** of some embodiments, it is originally difficult to generate the foreign matter X**1**, thereby overwhelmingly reducing a case where such foreign matter X**1** enters the bearing **42**, and facilitating improvement of reliability of the harmonic gear device **1**. Especially even when the harmonic gear device **1** is used for a long period of time, it is difficult for reduction of reliability thereof to occur, thereby facilitating long service life and high performance of the harmonic gear device **1**.

(155) Furthermore, in the harmonic gear device **1**, surface hardness of the internal tooth **21** is lower than that of the external tooth **31**. Therefore, a tooth width of the internal tooth **21** is greater than

that of the external tooth **31**, and in case that the external tooth **31** is received in a range of the internal tooth **21** in the tooth direction, sometimes a local recess (height difference) is generated at a portion of the tooth surface of the internal tooth **21** in the tooth direction D**1**, due to wear induced by contact between it and the external tooth **31** or the like. In such a state where the recess is generated, when a tooth contact position deviates in the tooth direction D**1** (a direction parallel to the rotation axis Ax**1**) due to certain jitter, an excessive load is acted at the meshing position of the internal tooth **21** and the external tooth **31**, resulting in abnormality of the harmonic gear device **1**. That is, even as to the external tooth **31** with relatively high surface hardness, since a corner portion such as an end portion in the tooth direction D**1** is in contact with the internal tooth **21** with a height difference, there is possibility of generating a hard foreign matter X**1** (with relatively high hardness) due to absence of the corner portion.

(156) In comparison, in the harmonic gear device **1** of some embodiments, the external tooth **31** protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21**. Therefore, in at least one side along the tooth direction D**1**, the internal tooth **21** with relatively low surface hardness is evenly worn due to tooth contact between it and the external tooth **31**, so that it is difficult to generate a local recess (height difference) at the tooth surface of the internal tooth **21**. Therefore, even though there is a situation where a tooth contact position deviates in the tooth direction D**1** due to certain jitter, occurrence of abnormality of the harmonic gear device **1** due to an excessive load acting at the meshing position of the internal tooth **21** and the external tooth **31** is easily suppressed. Therefore it is difficult to generate absence of a corner portion such as an end portion in the tooth direction D**1** of the external tooth **31**, as a result, it is difficult to generate a hard foreign matter X**1** (with relatively high hardness).

(157) Furthermore, as a basic structure of the harmonic gear device **1**, since there a large tooth number of the internal teeth **21** and the external teeth **31** meshing simultaneously, it has an advantage of dispersing a meshing load, and on the other hand, meshing loss is easy to increase due to a large sliding accompanied with meshing. Such meshing loss may become a primary cause of deterioration of starting performance of the harmonic gear device **1**, specially in a low temperature environment where the lubricant Lb**1** is easily hardened or the like. In comparison, according to the harmonic gear device **1** of some embodiments, the tooth-direction trimming portion **210** is arranged at an end portion in at least the opened surface **35** side along the tooth direction D**1** of the internal tooth **21**. Therefore, meshing of the internal tooth **21** and the external tooth **31** is reduced at the end portion at the opened surface **35** side with a very large amount of sliding in the tooth direction D**1**, so that the meshing loss may be reduced, and the power transmission efficiency may be improved. In this way, in the harmonic gear device **1** of some embodiments, not only its service life is prolonged, but also the starting performance of the harmonic gear device **1** for example in a low temperature environment where the lubricant Lb**1** is easily hardened, may be improved with improvement of the power transmission efficiency.

(6) Application Example

(158) Next, an application example of the harmonic gear device **1** and the actuator **100** of the embodiment is described with reference to FIG. **9**.

(159) FIG. **9** is a cross-sectional view showing an example of a robot **9** using the harmonic gear device **1** of some embodiments. The robot **9** is a horizontal multi-joint robot, i.e., a so-called Selective Compliance Assembly Robot Arm (SCARA) type robot.

(160) As shown in FIG. **9**, the robot **9** includes two harmonic gear devices **1**, and a connection rod **91**. The two harmonic gear devices **1** are arranged at joint parts at two positions of the robot **9** respectively. The connection rod **91** connects the joint parts at two positions to each other. In the example of FIG. **9**, the harmonic gear device **1** does not have a cup shape, instead, it is composed of a harmonic gear device with a top-hat shape. That is, in the harmonic gear device **1** shown in FIG. **9**, a flexible external gear **3** formed as a top-hat shape is used.

(7) Deformation Example

(161) The above embodiment is merely one of various implementations of some embodiments of the disclosure. As long as the above embodiment may achieve the purpose of some embodiments of the disclosure, various changes may be made thereto according to design or the like. Furthermore, drawings referred by some embodiments of the disclosure are schematic diagrams, and respective ratios of sizes and thicknesses of structural elements in the drawings do not necessarily reflect actual size ratios. Deformation examples of the above embodiment are listed below. The deformation examples described below may be applicable with proper combination.
(162) FIG. **10**A to FIG. **10**D show deformation examples of the above embodiment, which are equivalent to figures showing relationship between the internal teeth **21** and the external teeth **31** in a cross-sectional view of a line A**1**-A**1** of FIG. **5**A. Cross-sectional lines (oblique lines) of cross-sections are omitted in FIG. **10**A to FIG. **10**D.
(163) In a first deformation example shown in FIG. **10**A, the tooth-direction trimming portion (the first trimming portion) **210** of the internal tooth **21** is formed only on an end surface, at a side along a rotation direction R**1** of the flexible external gear **3**, of both end surfaces of the internal tooth **21** in the tooth-thickness direction D**2**. Furthermore, the tooth-direction trimming portion (the second trimming portion) **310** of the external tooth **31** is formed only on an end surface, at an opposite side along the rotation direction R**1** of the flexible external gear **3**, of both end surfaces of the external tooth **31** in the tooth-thickness direction D**2**. Therefore, during power transmission, tooth surfaces of the internal tooth **21** and the external tooth **31** formed with the tooth-direction trimming portions **210**, **310** are in contact with each other, and thus the same effect as that of the above embodiment may be expected.
(164) In a second deformation example shown in FIG. **10**B, the tooth-direction trimming portion (the first trimming portion) **210** of the internal tooth **21** is formed only on an end surface, at a side along a rotation direction R**1** of the flexible external gear **3**, of both end surfaces of the internal tooth **21** in the tooth-thickness direction D**2**. Furthermore, the tooth-direction trimming portion (the second trimming portion) **310** of the external tooth **31** is formed only on an end surface, at the side along the rotation direction R**1** of the flexible external gear **3**, of both end surfaces of the external tooth **31** in the tooth-thickness direction D**2**. In this structure, the same effect as that of the above embodiment may be expected by the tooth-direction trimming portions **210**, **310**.
(165) In a third deformation example shown in FIG. **10**C, only the internal tooth **21** of the internal tooth **21** and the external tooth **31** is provided with the tooth-direction trimming portion (the first trimming portion) **210**, and the external tooth **31** is not provided with a tooth-direction trimming portion. In this structure, the same effect as that of the above embodiment may be expected by the tooth-direction trimming portion **210**. As yet another example, there is a structure where only the external tooth **31** of the internal tooth **21** and the external tooth **31** may be provided with the tooth-direction trimming portion (the second trimming portion) **310**, and the internal tooth **21** is not provided with a tooth-direction trimming portion.
(166) In a fourth deformation example shown in FIG. **10**D, only an end portion of the internal tooth **21** in the tooth direction D**1** is provided with the tooth-direction trimming portion (the first trimming portion) **210**, and only an end portion of the external tooth **31** in the tooth direction D**1** is provided with the tooth-direction trimming portion (the second trimming portion) **310**. In the example of FIG. **10**D, the tooth-direction trimming portions **210**, **310** are arranged at end portions at the opened surface **35** side along the tooth direction D**1**. As yet another example, at least one of the tooth-direction trimming portions **210** or **310** may be arranged at an end portion at an opposite side of the opened surface **35** along the tooth direction D**1**.
(167) Structures of deformation examples shown in FIG. **10**A to FIG. **10**D may be applicable with proper combination. For example, With combination of the third deformation example and the fourth deformation example, only the internal tooth **21** of the internal tooth **21** and the external tooth **31** may be provided with the tooth-direction trimming portion **210**, and the tooth-direction trimming portion **210** is arranged at only an end portion in the tooth direction D**1** of the internal

tooth **21**.

(168) Furthermore, a case where tooth-shape trimming is performed on each of the internal tooth **21** and the external tooth **31**, is not a structure necessarily provided in the harmonic gear device **1**. For example, tooth-shape trimming may not be performed on at least one of the internal tooth **21** or the external tooth **31**.

(169) Furthermore, a structure where the internal tooth **21** has the tooth-direction trimming portion **210**, may be used separately from a structure where surface hardness of the internal tooth **21** is lower than that of the external tooth **31** and the external tooth **31** protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21**. That is, even when the structure where surface hardness of the internal tooth **21** is lower than that of the external tooth **31** and the external tooth **31** protrudes toward at least one side along the tooth direction D**1** relative to the internal tooth **21** is used separately, it is difficult to generate the foreign matter X**1** due to absence, or wear, or the like induced by contact between the internal tooth **21** and the external tooth **31**. Therefore, even when the internal tooth **21** does not have the tooth-direction trimming portion **210**, it is possible to provide the harmonic gear device **1** with high reliability. On the other hand, even when the structure where the internal tooth **21** has the tooth-direction trimming portion **210** is used separately, it is difficult to generate the foreign matter X**1** due to absence, or wear, or the like induced by contact between the internal tooth **21** and the external tooth **31**. Therefore, even when the external tooth **31** does not protrude toward at least one side along the tooth direction D**1** relative to the internal tooth **21**, it is possible to provide the harmonic gear device **1** with high reliability.

(170) Furthermore, a structure where the lubricant Lb**1** is maintained in the gap between the tapered surface **302** and the outer peripheral surface **424** of the wave generator **4**, may be used separately. That is, even when the internal tooth **21** does not have the tooth-direction trimming portion **210** and the external tooth **31** does not protrude toward at least one side along the tooth direction D**1** relative to the internal tooth **21**, the gap between the tapered surface **302** and the outer peripheral surface **424** of the wave generator **4** may maintain the lubricant Lb**1**.

(171) Furthermore, the harmonic gear device **1** is not limited to the cup shape described in the above embodiments, for example, it may have a top-hat shape, a ring shape, a differential shape, a flat shape (flattening shape), or a cover housing shape, or the like. For example, the harmonic gear device **1** with a top-hat shape as shown in FIG. **9** has a cylindrical flexible external gear **3** with an opened surface **35** at a side along the tooth direction D**1**, like the harmonic gear device with a cup shape. That is, an end portion, at a side of the rotation axis Ax**1**, of the flexible external gear **3** with a top-hat shape has a flange portion, and an end portion at an opposite side of the flange portion has the opened surface **35**. Even as to the flexible external gear **3** with a top-hat shape, an end portion at the opened surface **35** side has the external tooth **31**, and the wave generator **4** is embedded in it.

(172) Furthermore, structure of the actuator **100** is not limited to structures shown in the above embodiment, and may be appropriately changed. For example, the connection structure of the input portion **103** and the cam **41** is not limited to a spline connection structure, and may use an Euclidean joint or the like. By using the Euclidean joint as the connection structure of the input portion **103** and the cam **41**, eccentricity between the wave generator **4** (the cam **41**) and the rotation axis Ax**1** at the input side may be counteracted, and eccentricity between the rigid internal gear **2** and the flexible external gear **3** may be counteracted. Furthermore, the cam **41** may be unable to move along the rotation axis Ax**1** relative to the input portion **103**.

(173) Furthermore, the application example of the harmonic gear device **1** and the actuator **100** of the embodiment is not limited to the above horizontal multi-joint robot, for example, may also be applicable to industrial robots other than the horizontal multi-joint robot, or robots other than industrial robots, or the like. As an example, in industrial robots other than the horizontal multi-joint robot, there is a vertical multi-joint robot, or a parallel link-type robot, or the like. As an example, in robots other than industrial robots, there is a home robot, a nursing robot, or a medical robot, or the like.

(174) Furthermore, the bearing **42** is not limited to a deep groove ball bearing, for example, it may be an angular contact ball bearing or the like. Furthermore, the bearing **42** is not limited to a ball bearing, for example, it may be a roller bearing such as a cylindrical roller bearing, a needle roller bearing, or a conical roller bearing, or the like in which the rolling body **423** is composed of a non-spherical “roller”.
(175) Furthermore, materials of structural elements of the harmonic gear device **1** or the actuator **100** are not limited to metal, for example, they may be resins such as an engineering plastic or the like.
(176) Furthermore, the lubricant Lb**1** is not limited to a liquid substance such as lubrication oil (oil) or the like, and may be a gel-like substance such as a lubrication grease or the like.
(177) Furthermore, protrusion amounts L**1**, L**2** by which the external tooth **31** protrudes from the internal tooth **21** are not limited to be substantially the same at both sides along the tooth direction D**1**. For example, the protrusion amount L**1** at one side along the tooth direction D**1** (the input side of the rotation axis Ax**1**) may be greater than the protrusion amount L**2** at the other side along the tooth direction D**1** (the output side of the rotation axis Ax**1**). On the contrary, the protrusion amount L**1** at one side along the tooth direction D**1** (the input side of the rotation axis Ax**1**) may be less than the protrusion amount L**2** at the other side along the tooth direction D**1** (the output side of the rotation axis Ax**1**).
(178) Furthermore, a case where the trimming amount Q**2** of the second trimming portion **310** is less than the trimming amount Q**1** of the first trimming portion **210**, is not a structure necessarily provided in the harmonic gear device **1**. For example, the trimming amount Q**2** of the second trimming portion **310** may be equal to the trimming amount Q**1** of the first trimming portion **210** (Q**1**=Q**2**), or the trimming amount Q**2** of the second trimming portion **310** may be greater than the trimming amount Q**1** of the first trimming portion **210** (Q**1**<Q**2**).
(179) As shown in FIG. **11**A to FIG. **12**C, the harmonic gear device **1**A of another embodiment is different from the harmonic gear device **1** of the above embodiment in forming the tooth-direction trimming portion (the first trimming portion) **210** of the internal tooth **21** by an edge trimming process. Hereinafter, the same structure as that described in the above embodiment is marked with the same reference numeral, and descriptions thereof are omitted appropriately.
(180) FIG. **11**A is a cross-sectional view in view of the internal tooth **21** and the external tooth **31**. FIG. **11**B is a cross-sectional view of a line A**1**-A**1** of FIG. **11**A. FIG. **12**A is a cross-sectional view of a line n**1**-n**1** of FIG. **11**A. FIG. **12**B is a cross-sectional view of a line B**2**-B**2** of FIG. **11**A. FIG. **12**C is a cross-sectional view of a line B**3**-B**3** of FIG. **11**A.
(181) The tooth-direction trimming portion (the first trimming portion) **210** of the internal tooth **21** is formed by drum-shaped trimming in the above embodiment, while formed by an edge trimming process in this embodiment. That is, the first trimming portion **210** is formed by processing the internal tooth **21** so that a central portion in the tooth direction D**1** still has a regular tooth-direction shape and only both end portions in the tooth direction D**1** are processed into tapered shapes, in a manner that the central portion in the tooth direction D**1** becomes a protrusion. Similarly, in some embodiments, the tooth-direction trimming portion (the second trimming portion) **310** of the external tooth **31** is also formed by an edge trimming process. In this way, the first trimming portion **210** and the second trimming portion **310** are based on the same type of tooth-direction trimming (here, the edge trimming process).
(182) In some embodiments, as shown in FIG. **11**A and FIG. **11**B, tooth-direction trimming composed of the edge trimming process is performed on the tooth bottom **212**, the tooth crest **213**, and both end surfaces in the tooth-thickness direction D**2** of the internal tooth **21**. The above principle is also applicable to the external tooth **31**, tooth-direction trimming composed of the edge trimming process is performed on the tooth bottom **312**, the tooth crest **313**, and both end surfaces in the tooth-thickness direction D**2** of the external tooth **31**.
(183) According to the tooth-direction trimming portions **210**, **310** as described above, as shown in

FIG. **11**B, tooth thicknesses of each of the internal tooth **21** and the external tooth **31** is maximal at a central portion in the tooth direction D**1**, and gradually reduces toward both end portions in the tooth direction D**1**, after a starting point of the edge trimming process. Therefore, at the meshing position of the internal tooth **21** and the external tooth **31**, a gap between the internal tooth **21** and the external tooth **31** is minimal at a central portion (non-trimming portion) on which tooth-direction trimming (here, the edge trimming process) is not performed in the tooth direction D**1**.
(184) In summary, in the examples of FIG. **12**A to FIG. **12**C, a gap G**1** between the internal tooth **21** and the external tooth **31** is minimal at a cross section of the starting point of the edge trimming process in the tooth direction D**1**, i.e., in FIG. **12**A. That is, the gap G**1** between the internal tooth **21** and the external tooth **31** becomes larger toward an end portion in the tooth direction D**1** (the output side of the rotation axis Ax**1**), according to a sequence of FIG. **12**B and FIG. **12**C. In this way, by tooth-direction trimming, the farther away from the central portion in the tooth direction D**1**, the greater the tooth surface shifting in a negative direction gradually, therefore, the gap G**1** between the internal tooth **21** and the external tooth **31** becomes larger.
(185) As a deformation example of some embodiments, for example, the first trimming portion **210** and the second trimming portion **310** may include different types of tooth-direction trimming, for example, the first trimming portion **210** is subject to the edge trimming process while the second trimming portion **310** is subject to the drum-shaped trimming. On the contrary, the first trimming portion **210** is subject to the drum-shaped trimming while the second trimming portion **310** is subject to the edge trimming process.
(186) The structure (including the deformation example) of the embodiment may be suitable for appropriate combination with the structure (including the deformation example) described in the above embodiment.
(187) As shown in FIG. **13**, the harmonic gear device **1**B of yet another embodiment is different from the harmonic gear device **1** of the above embodiment in the external tooth **31** protruding toward only one side along the tooth direction D**1** relative to the internal tooth **21**. FIG. **13** is a cross-sectional view in view of the internal tooth **21** and the external tooth **31**, with cross-sectional lines (oblique lines) of the cross section omitted. Hereinafter, the same structure as that described in the above embodiment is marked with the same reference numeral, and descriptions thereof are omitted appropriately.
(188) That is, in the above embodiment, the external tooth **31** protrudes toward both sides along the tooth direction D**1** (input and output sides of the rotation axis Ax**1**) relative to the internal tooth **21**. In comparison, in some embodiments, the external tooth **31** protrudes toward only one side along the tooth direction D**1** relative to the internal tooth **21**. Especially in some embodiments, the external tooth **31** protrudes toward the opened surface **35** side along the tooth direction D**1**, i.e., toward the input side of the rotation axis Ax**1**, relative to the internal tooth **21**. That is, in some embodiments, the external tooth **31** protrudes toward the opened surface **35** side along the tooth direction D**1** (the input side of the rotation axis Ax**1**) relative to the internal tooth **21**, and does not protrude toward an opposite side of the opened surface **35** along the tooth direction D**1** (the output side of the rotation axis Ax**1**).
(189) Here, in a state where the flexible external gear **3** is elastically deformed, an end portion at the opened surface **35** side of the flexible external gear **3** in the direction of the rotation axis Ax**1** is deformed greater than deformation of an end portion at the bottom portion **322** side, to form a shape closer to an elliptical shape. As in some embodiments, the external tooth **31** protrudes from the internal tooth **21** at the opened surface **35** side along the tooth direction D**1**, thereby avoiding a corner portion at a front end of the external tooth **31** inclined due to such tapered surface **302** from contacting the internal tooth **21**. Therefore, according to the structure of some embodiments, it is difficult to generate a local recess (height difference) at the tooth surface of the internal tooth **21**, at the end portion at the opened surface **35** side along the tooth direction D**1** where stress concentration is generated very easily due to deformation of the external tooth **31**.

(190) As a deformation example of some embodiments, the external tooth **31** may be equivalent to the internal tooth **21**, and only protrudes toward the opposite side of the opened surface **35** along the tooth direction D**1**, i.e., toward the output side of the rotation axis Ax**1**. In this case, the external tooth **31** does not protrude toward the opened surface **35** side along the tooth direction D**1** (the input side of the rotation axis Ax**1**) relative to the internal tooth **21**.
(191) The structure (including the deformation example) of the embodiment may be suitable for appropriate combination with the structure (including the deformation example) described in the above embodiment or another embodiment.
(192) (Summary)
(193) As explained above, a harmonic gear device (**1**, **1**A, **1**B) with a first configuration includes a rigid internal gear (**2**), a flexible external gear (**3**) and a wave generator (**4**). The rigid internal gear (**2**) is an annular component having internal teeth (**21**). The flexible external gear (**3**) is an annular component having external teeth (**31**) and arranged on an inner side of the rigid internal gear (**2**). The wave generator (**4**) is arranged on an inner side of the flexible external gear (**3**) and deflects the flexible external gear (**3**). The harmonic gear device (**1**, **1**A, **1**B) deforms the flexible external gear (**3**) along with rotation of the wave generator (**4**) with a rotation axis (Ax**1**) as a center, such that a part of the external teeth (**31**) is meshed with a part of the internal teeth (**21**), and thus the flexible external gear (**3**) relatively rotates relative to the rigid internal gear (**2**) according to a difference between tooth numbers of the flexible external gear (**3**) and the rigid internal gear (**2**). The internal tooth (**21**) has a tooth-direction trimming portion (**210**) at an end portion in at least one side along a tooth direction (D**1**) of the internal tooth (**21**).
(194) According to this configuration, an “avoidance portion” is formed between the tooth-direction trimming portion (**210**) of the internal tooth (**21**) and the external tooth (**31**), therefore, at an end portion in at least one side along the tooth direction (D**1**) of the internal tooth (**21**), it is difficult to generate stress concentration induced by excessive tooth contact between the internal tooth (**21**) and the external tooth (**31**). Especially in the harmonic gear device (**1**, **1**A, **1***i*), the wave generator (**4**) deflects the flexible external gear (**3**), therefore deformation of the external tooth (**31**) such as torsion, deflection (inclination), or the like is sometimes generated with respect to the rotation axis (Ax**1**). Therefore, although stress concentration induced by deformation of the external tooth (**31**) is easily generated at an end portion in at least one side along the tooth direction (D**1**) of the internal tooth (**21**), it is difficult to generate such stress concentration through the tooth-direction trimming portion (**210**). Therefore it is difficult to generate the foreign matter (X**1**) due to absence, or wear, or the like induced by contact between the internal tooth (**21**) and the external tooth (**31**), which may provide the harmonic gear device (**1**) with high reliability. Furthermore, since the tooth-direction trimming portion (**210**) is arranged on the rigid internal gear (**2**), tooth-direction trimming is unnecessary for the flexible external gear (**3**), or a trimming amount of the flexible external gear (**3**) may be reduced, and reduction of strength of the flexible external gear (**3**) induced by performing tooth-direction trimming on the flexible external gear (**3**) with flexibility is easily suppressed.
(195) In a harmonic gear device (**1**, **1**A, **1**B) with a second configuration, based on the first configuration, the internal tooth (**21**) has a first trimming portion (**210**) used as the tooth-direction trimming portion (**210**). The external tooth (**31**) has a second trimming portion (**310**) which is a tooth-direction trimming portion (**310**) different from the first trimming portion (**210**).
(196) According to this configuration, since both the internal tooth (**21**) and the external tooth (**31**) have tooth-direction trimming portions (**210**, **310**) respectively, tooth contact of the internal tooth (**21**) and the external tooth (**31**) may be improved.
(197) In a harmonic gear device (**1**, **1**A, **1**B) with a third configuration, based on the second configuration, a trimming amount (Q**2**) of the second trimming portion (**310**) is less than that (Q**1**) of the first trimming portion (**210**).
(198) According to this configuration, tolerance of the bending stress required by the external tooth

(**31**) of the flexible external gear (**3**) with elastic deformation (deflection) may be easily achieved by suppressing the trimming amount (Q**2**) of the external tooth (**31**) to be small.
(199) In a harmonic gear device (**1**, **1**A, **1**B) with a fourth configuration, based on the second or third configuration, the first trimming portion (**210**) and the second trimming portion (**310**) include the same type of trimming.
(200) According to this configuration, the first trimming portion (**210**) and the second trimming portion (**310**) are easy to be processed.
(201) In a harmonic gear device (**1**, **1**A, **1**B) with a fifth configuration, based on any one of the first to fourth configurations, an inner peripheral surface (**301**) of the flexible external gear (**3**) has a tapered surface (**302**) at a position opposite to an outer peripheral surface (**442**) of the wave generator (**4**). The tapered surface (**302**) increases a gap between the tapered surface (**302**) and the outer peripheral surface (**442**) of the wave generator (**4**) in a direction along the rotation axis (Ax**1**). A lubricant (Lb**1**) is maintained in the gap between the tapered surface (**302**) and the outer peripheral surface (**442**) of the wave generator (**4**).
(202) According to this configuration, it is easy to suppress entry of the foreign matter (X**1**) through the gap between the tapered surface (**302**) and the outer peripheral surface (**442**) of the wave generator (**4**).
(203) In a harmonic gear device (**1**, **1**A, **1**B) with a sixth configuration, based on the fifth configuration, an inclination angle (θ**1**) of the tapered surface (**302**) with respect to the rotation axis (Ax**1**) is less than 5 degrees.
(204) According to this configuration, the gap between the tapered surface (**302**) and the outer peripheral surface (**442**) of the wave generator (**4**) may be a small gap, to maintain the lubricant (Lb**1**) for example by a capillary phenomenon.
(205) In a harmonic gear device (**1**, **1**A, **1**B) with a seventh configuration, based on any one of the first to sixth configurations, the tooth-direction trimming portion (**210**) includes an inclined surface at at least one of a tooth bottom (**212**), a tooth crest (**213**), or an end surface in a tooth-thickness direction (D**2**) of the internal tooth (**21**). The inclined surface is inclined with respect to the tooth direction (D**1**).
(206) According to this configuration, tooth contact may be improved by performing tooth-direction trimming on at least one of the tooth bottom (**212**), the tooth crest (**213**), or an end surface in the tooth-thickness direction (D**2**) of the internal tooth (**21**).
(207) In a harmonic gear device (**1**, **1**A, **1**B) with an eighth configuration, based on the seventh configuration, the tooth-direction trimming portion (**210**) includes the inclined surface at least at the tooth bottom (**212**).
(208) According to this configuration, it is easy to avoid contact of the tooth crest (**313**) of the external tooth (**31**) and the tooth bottom (**212**) of the internal tooth (**21**).
(209) In a harmonic gear device (**1**, **1**A, **1**B) with a ninth configuration, based on any one of the first to eighth configurations, the flexible external gear (**3**) has a cylindrical shape with an opened surface (**35**) at a side along the tooth direction (D**1**). The tooth-direction trimming portion (**210**) is arranged at an end portion in at least the opened surface (**35**) side of the internal tooth (**21**) along the tooth direction (D**1**).
(210) According to this configuration, it may be difficult to generate stress concentration through the tooth-direction trimming portion (**210**), at the end portion at the opened surface (**35**) side along the tooth direction (D**1**) where stress concentration is generated very easily due to deformation of the external tooth (**31**).
(211) An actuator (**100**) with a tenth configuration includes the harmonic gear device (**1**, **1**A, **1**B) with any one of the first to ninth configurations, a driving source (**101**) and an output portion (**102**). The driving source (**101**) is configured to rotate the wave generator (**4**). The output portion (**102**) is configured to take a rotation force of one of the rigid internal gear (**2**) or the flexible external gear (**3**) out as output.

(212) According to this configuration, it is difficult to generate the foreign matter (X**1**) due to absence, or wear, or the like induced by contact between the internal tooth (**21**) and the external tooth (**31**), which may provide the actuator (**100**) with high reliability.
(213) Furthermore, the structure of the harmonic gear device (**1**, **1**A, **1**B) with a fifth configuration may be used separately, regardless of presence or absence of the tooth-direction trimming portion (**210**) with a first configuration. That is, the harmonic gear arrangement (**1**, **1**A, **1**B) includes a rigid internal gear (**2**), a flexible external gear (**3**) and a wave generator (**4**). The rigid internal gear (**2**) is an annular component having internal teeth (**21**). The flexible external gear (**3**) is an annular component having external teeth (**31**) and arranged on an inner side of the rigid internal gear (**2**). The wave generator (**4**) is arranged on an inner side of the flexible external gear (**3**) and deflects the flexible external gear (**3**). The harmonic gear device (**1**, **1**A, **1**B) deforms the flexible external gear (**3**) along with rotation of the wave generator (**4**) with a rotation axis (Ax**1**) as a center, such that a part of the external teeth (**31**) is meshed with a part of the internal teeth (**21**), allowing the flexible external gear (**3**) to relatively rotate relative to the rigid internal gear (**2**) according to a difference between tooth numbers of the flexible external gear (**3**) and the rigid internal gear (**2**). Here an inner peripheral surface (**301**) of the flexible external gear (**3**) has a tapered surface (**302**) at a position opposite to an outer peripheral surface (**442**) of the wave generator (**4**). The tapered surface (**302**) increases a gap between the tapered surface (**302**) and the outer peripheral surface (**442**) of the wave generator (**4**) in a direction along the rotation axis (Ax**1**). A lubricant (Lb**1**) is maintained in the gap between the tapered surface (**302**) and the outer peripheral surface (**442**) of the wave generator (**4**).
(214) With regard to structures of the second to ninth configurations, structures which are unnecessary to be provided for the harmonic gear device (**1**, **1**A, **1**B) may be omitted appropriately.
EXPLANATIONS OF REFERENCE NUMERALS
(215) **1**, **1**A,**1**B Harmonic gear device **2** Rigid internal gear **3** Flexible external gear **4** Wave generator **21** Internal tooth **31** External tooth **35** Opened surface **100** Actuator **101** Driving source **102** Output portion **210** Tooth-direction trimming portion (First trimming portion) **212** Tooth bottom **213** Tooth crest **301** Inner peripheral surface (of Flexible external gear) **302** Tapered surface **310** Tooth-direction trimming portion (Second trimming portion) **424** Outer peripheral surface (of Wave generator) Ax**1** Rotation axis D**1** Tooth direction D**2** Tooth-thickness direction Lb**1** Lubricant Q**1** Trimming amount of First trimming portion Q**2** Trimming amount of Second trimming portion θ**1** Inclination angle
INDUSTRIAL APPLICABILITY
(216) According to some embodiments of the disclosure, a harmonic gear device and an actuator with high reliability may be provided.

Claims

1. A harmonic gear device, comprising: a rigid internal gear with an annular shape, having a first number of internal teeth; a flexible external gear with an annular shape, having a second number of external teeth and arranged on an inner side of the rigid internal gear; and a wave generator, arranged on an inner side of the flexible external gear and configured to deflect the flexible external gear, wherein the harmonic gear device is configured to deform the flexible external gear by a rotation of the wave generator along a rotation axis, such that a portion of the external teeth is meshed with a portion of the internal teeth, and the flexible external gear is configured to rotate relative to the rigid internal gear in accordance with a difference between the first number of internal teeth and the second number of external teeth, and wherein an internal tooth of the first number of internal teeth has a tooth-direction trimming portion at an end portion on at least one side along a tooth direction of the internal tooth, wherein an inner peripheral surface of the flexible external gear has a tapered surface at a position opposite to an outer peripheral surface of the wave

generator, and the tapered surface increases a gap between the tapered surface and the outer peripheral surface of the wave generator in a direction along the rotation axis, and a lubricant is maintained in the gap between the tapered surface and the outer peripheral surface of the wave generator, wherein the tapered surface is a tapered surface of an external tooth of the second number of external teeth, and the external tooth protrudes from the internal tooth.

2. The harmonic gear device of claim 1, wherein the tooth-direction trimming portion includes a first trimming portion of the internal tooth, and the external tooth of the second number of external teeth includes a second trimming portion which is a tooth-direction trimming portion different from the first trimming portion.

3. The harmonic gear device of claim 2, wherein a trimming amount of the second trimming portion is less than that of the first trimming portion.

4. The harmonic gear device of claim 2, wherein the first trimming portion and the second trimming portion are of the same type.

5. The harmonic gear device of claim 1, wherein an inclination angle of the tapered surface with respect to the rotation axis is less than 5 degrees.

6. The harmonic gear device of claim 1, wherein the tooth-direction trimming portion comprises an inclined surface inclined with respect to the tooth direction, in at least one of a tooth bottom, a tooth crest, or an end surface in a tooth-thickness direction of the internal tooth.

7. The harmonic gear device of claim 6, wherein the tooth-direction trimming portion comprises the inclined surface at least at the tooth bottom.

8. The harmonic gear device of claim 1, wherein the flexible external gear has a cylindrical shape with an opened surface at a side along the tooth direction, the tooth-direction trimming portion is arranged at an end portion of at least the opened surface at the side along the tooth direction of the internal tooth.

9. An actuator, comprising: the harmonic gear device of claim 1; a driving source configured to rotate the wave generator; and an output portion configured to provide a rotation force of one of the rigid internal gear or the flexible external gear out as an output.

10. The actuator of claim 9, wherein the tooth-direction trimming portion includes a first trimming portion of the internal tooth, and the external tooth of the second number of external teeth includes a second trimming portion which is a tooth-direction trimming portion different from the first trimming portion.

11. The actuator of claim 10, wherein a trimming amount of the second trimming portion is less than that of the first trimming portion.

12. The actuator of claim 10, wherein the first trimming portion and the second trimming portion are of the same type.

13. The actuator of claim 9, wherein an inclination angle of the tapered surface with respect to the rotation axis is less than 5 degrees.

14. The actuator of claim 9, wherein the tooth-direction trimming portion comprises an inclined surface inclined with respect to the tooth direction, in at least one of a tooth bottom, a tooth crest, or an end surface in a tooth-thickness direction of the internal tooth.

15. The actuator of claim 14, wherein the tooth-direction trimming portion comprises the inclined surface at least at the tooth bottom.

16. The actuator of claim 9, wherein the flexible external gear has a cylindrical shape with an opened surface at a side along the tooth direction, the tooth-direction trimming portion is arranged at an end portion of at least the opened surface side along the tooth direction of the internal tooth.